Supplementary Material for LIVE: An LLM-assisted Multi-Image Visual Hallucination Evaluation Benchmark

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CCS Concepts

• **Computing methodologies** → **Scene understanding**; *Activity recognition and understanding*; **Language resources**; *Natural language generation.*

Keywords

Large vision-language models, Multi-image hallucination, Multi-granularity evaluation

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1 Empirical Selection of the Number of Clusters

To construct diverse yet semantically coherent multi-image scenarios, we first cluster the CLIP features of all COCO val2014 images using K-means. We vary the number of clusters K from 12 to 5,000 and record three statistics: (1) the within-cluster sum of squares (WCSS), (2) the number of clusters containing at least 4 images, and (3) the number of clusters containing at least 50 images.

As shown in Figure 1, WCSS decreases rapidly when increasing K in the low-K regime and starts to decrease slowly and smoothly around $K = 1{,}000$, indicating diminishing returns in intra-cluster compactness beyond this point. At the same time, the number of clusters with at least 4 images keeps growing, providing a large pool of small but non-trivial clusters that support diverse multi-image compositions. In contrast, the number of high-density clusters (size ≥ 50) gradually drops as K increases, since images are split into finer groups. We therefore choose $K = 1{,}000$ as a compromise point where WCSS is already substantially reduced, while we still retain a sufficient number of large clusters (size ≥ 50) for reliable sampling and a rich set of medium-sized clusters (size ≥ 4) to ensure diversity. This empirical choice is used throughout the main experiments.

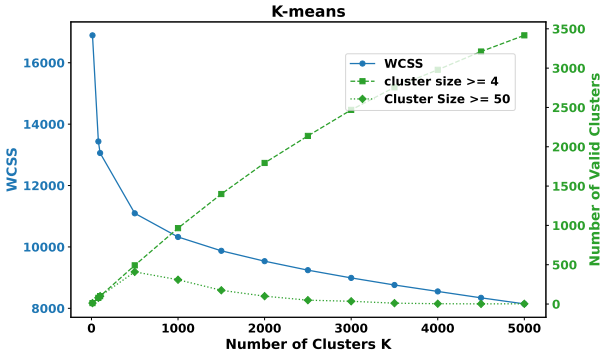

Figure 1: K-means analysis for different numbers of clusters K.

2 Prompting Templates for LLM-Assisted Counterfactual Generation

We provide the prompting templates used in the LLM-assisted counterfactual generation stage. Placeholders such as {captions_str}, {seed_object}, {class_type}, and {seed_relationship} are instantiated with the scene-graph-derived content for each multi-image scenario.

Object-level counterfactual generation.

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You are a creative assistant tasked with generating conflicting information.
Background Context (from image captions): {captions_str}
Seed Object: {seed_object}
Your task is to generate a *new object* that must satisfy the following strict rules:
1. The new object must be conceptually related to the Seed Object, or be its direct antonym / conflicting item (e.g., if the seed is "horse", the new object could be "motorcycle" or "zebra").
2. The new object **must** conflict with the Background Context. For example, if the context is about a "horse" pulling a "carriage", generating "motorcycle" is a good conflict.
3. The new object **must not** appear in any of the Background Context descriptions.
4. The new object **must not** be similar to anything in the Background Context descriptions.
5. Provide a brief reason explaining why this new object is conflicting.
Output Format (strictly follow):
Object: `<your new generated object>`
Reason: `<your reason>`

Attribute-level counterfactual generation.

You are a creative assistant tasked with generating conflicting information.
Background Context (from image captions): {captions_str}
Seed Attribute Class: {class_type}
Seed Attribute: {seed_attribute}
Your task is to generate a *new attribute* for the same object, following these strict rules:
1. The new attribute must be from the same class ({class_type}). For example, {instruction}.
2. The new attribute **must** conflict with the Background Context. For example, if the context describes "black horses" and the seed is "horses are black", a good conflict would be "horses are white".
3. The new attribute **must not** be semantically similar to, or mentioned in, the Background Context.
4. Provide a brief reason explaining why this new attribute is conflicting with the context.
Output Format (strictly follow):
Attribute: `<your new generated attribute>`
Reason: `<your reason>`

Relationship-level counterfactual generation.

You are a creative assistant tasked with generating conflicting information.
Background Context (from image captions): {captions_str}
Seed Relationship Class: {relationship_class}
Seed Relationship: {seed_relationship}
Your task is to generate a *new relationship* that must satisfy the following strict rules:
1. The new relationship must be a conflicting {relationship_class}.
2. The new relationship **must** conflict with and be semantically dissimilar to the Background Context.
3. The new relationship **must not** be semantically similar to, or mentioned in, the Background Context.
4. Provide a brief reason explaining why this new relationship is conflicting with the context. Output Format (strictly follow):
Relationship: `<your new generated relationship>`
Reason: `<your reason>`

3 Detailed Ablation Study Results

We present the comprehensive, full-scale results of our ablation study in Table 1. This table extends our primary findings by detailing the performance of all evaluated Vision-Language Models (VLMs)—ranging from proprietary models like GPT-4o to various open-weight models (e.g., Qwen-VL, InternVL, MiniCPM, Mantis, and LLaVA).

To thoroughly evaluate model robustness and context utilization, the results are divided into two primary testing environments: *Uniform Image Context* and *Diverse Image Context*. Within each scenario, we systematically analyze two key dimensions:

- **Ablation of Image Quantity Effects:** We observe how incrementally reducing the number of targeted images (e.g., from 4 to 3, and down to 2) impacts the F1 score and Accuracy of each model.
- **Ablation of Image Order Effects:** We assess the sensitivity of the models to the sequence of visual inputs. The † symbol denotes settings where the targeted images were randomly shuffled to test order dependency.

For ease of reading, performance fluctuations are tracked against the baseline configuration of each task, with ↑ and ↓ indicating an increase or decrease in the respective metrics.

4 Detailed Multi-Granularity Evaluation Results under Different Temperatures

In this section, we provide the comprehensive performance breakdowns of all evaluated LVLMs on the LIVE benchmark, supplementing the aggregated MAP results presented in Section 4.2. To rigorously assess the robustness of our benchmark and the stability of the models' behaviors, we conduct evaluations across three different sampling temperatures: 0 (greedy decoding for deterministic outputs, used in our main experiments), 0.7, and 1.0.

While varying the temperature introduces expected fluctuations in absolute metric scores—reflecting the inherent trade-off between response diversity and factual grounding—the core observations remain highly consistent. Specifically, the severe performance degradation in Diverse Image Context (DIC) scenarios and the pronounced vulnerabilities in high-level reasoning tasks (e.g.,

Table 1: Ablation study results.

Task	GPT-4o		Qwen3-VL-32B		InternVL-2.5-26B		Qwen3-VL-8B		InternVL-2.5-8B		MiniCPM-V4.5		MiniCPM-V2.6		Mantis-8B		Qwen2.5-VL-7B		LLaVA-OV		LLaVA-IL	
Metrics	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc
Scenario 1: Uniform Image Context																						
Ablation of Image Quantity Effects																						
TI_1@4	0.945	0.946	0.935	0.939	0.870	0.885	0.909	0.919	0.801	0.838	0.944	0.946	0.901	0.907	0.859	0.845	0.830	0.858	0.884	0.893	0.884	0.877
TI_1@3	0.949↑	0.951↑	0.938↑	0.943↑	0.920↑	0.925↑	0.912↑	0.921↑	0.844↑	0.866↑	0.947↑	0.949↑	0.912↑	0.917↑	0.876↑	0.864↑	0.832↑	0.858	0.908↑	0.913↑	0.893↑	0.888↑
TI_1@2	0.940↓	0.943↓	0.938↑	0.942↑	0.937↑	0.941↑	0.911↑	0.921↑	0.897↑	0.906↑	0.956↑	0.957↑	0.912↑	0.916↑	0.890↑	0.881↑	0.841↑	0.866↑	0.934↑	0.936↑	0.902↑	0.896↑
TI_2@4	0.839	0.866	0.842	0.869	0.881	0.888	0.783	0.828	0.777	0.817	0.811	0.839	0.624	0.734	0.823	0.833	0.467	0.661	0.841	0.861	0.883	0.871
TI_2@3	0.853↑	0.877↑	0.849↑	0.871↑	0.890↑	0.897↑	0.783	0.827↓	0.796↑	0.831↑	0.814↑	0.841↑	0.625↑	0.736↑	0.845↑	0.852↑	0.433↓	0.645↓	0.858↑	0.874↑	0.892↑	0.885↑
Ablation of Image Order Effects																						
TI_3	0.843	0.869	0.878	0.894	0.848	0.867	0.823	0.855	0.852	0.872	0.860	0.875	0.571	0.708	0.842	0.852	0.431	0.649	0.875	0.889	0.904	0.898
$TI_3^{\dagger}$	0.842↓	0.869	0.858↓	0.880↓	0.850↑	0.869↑	0.814↓	0.848↓	0.856↑	0.875↑	0.848↓	0.865↓	0.541↓	0.692↓	0.843↑	0.853↑	0.425↓	0.644↓	0.883↑	0.895↑	0.904	0.899↑
TI_4	0.889	0.902	0.927	0.932	0.863	0.877	0.900	0.910	0.881	0.893	0.880	0.891	0.731	0.793	0.867	0.875	0.455	0.659	0.894	0.903	0.915	0.912
$TI_4^{\dagger}$	0.885↓	0.899↓	0.910↓	0.917↓	0.866↑	0.881↑	0.908↑	0.916↑	0.883↑	0.895↑	0.867↓	0.881↓	0.719↓	0.786↓	0.873↑	0.881↑	0.439↓	0.650↓	0.903↑	0.911↑	0.910↓	0.906↓
Scenario 2: Diverse Image Context																						
Ablation of Image Quantity Effects																						
TI_1@4	0.953	0.954	0.924	0.928	0.871	0.887	0.907	0.917	0.800	0.835	0.930	0.933	0.908	0.913	0.853	0.842	0.836	0.861	0.893	0.896	0.877	0.870
TI_1@3	0.936↓	0.938↓	0.925↑	0.929↑	0.926↑	0.929↑	0.911↑	0.919↑	0.846↑	0.868↑	0.936↑	0.939↑	0.913↑	0.917↑	0.876↑	0.869↑	0.837↑	0.863↑	0.921↑	0.923↑	0.886↑	0.882↑
TI_1@2	0.935↓	0.938↓	0.933↑	0.936↑	0.937↑	0.939↑	0.898↓	0.910↓	0.893↑	0.906↑	0.944↑	0.946↑	0.915↑	0.919↑	0.898↑	0.890↑	0.855↑	0.875↑	0.933↑	0.933↑	0.907↑	0.901↑
TI_2@4	0.792	0.838	0.790	0.835	0.784	0.816	0.643	0.747	0.659	0.750	0.698	0.768	0.427	0.646	0.752	0.787	0.242	0.572	0.737	0.792	0.874	0.869
TI_2@3	0.809↑	0.849↑	0.802↑	0.841↑	0.801↑	0.832↑	0.642↓	0.746↓	0.647↓	0.744↓	0.689↓	0.766↓	0.397↓	0.633↓	0.779↑	0.806↑	0.203↓	0.556↓	0.748↑	0.800↑	0.881↑	0.879↑
Ablation of Image Order Effects																						
TI_3	0.796	0.841	0.833	0.863	0.720	0.780	0.712	0.782	0.717	0.779	0.801	0.832	0.324	0.601	0.776	0.803	0.189	0.556	0.774	0.816	0.874	0.875
$TI_3^{\dagger}$	0.795↓	0.843↑	0.837↑	0.863	0.708↓	0.773↓	0.721↑	0.788↑	0.721↑	0.782↑	0.800↓	0.832	0.324	0.601	0.773↓	0.801↓	0.161↓	0.547↓	0.774	0.814↓	0.875↑	0.876↑
TI_4	0.847	0.874	0.894	0.903	0.756	0.801	0.829	0.853	0.776	0.814	0.799	0.826	0.561	0.704	0.788	0.810	0.216	0.566	0.838	0.859	0.881	0.880
$TI_4^{\dagger}$	0.857↑	0.879↑	0.878↓	0.890↓	0.740↓	0.793↓	0.831↑	0.856↑	0.783↑	0.821↑	0.810↑	0.837↑	0.556↓	0.702↓	0.772↓	0.800↓	0.212↓	0.563↓	0.838	0.857↓	0.882↑	0.881↑

Note: † denotes random shuffling of targeted images. ↑ /↓ denote an increase/decrease versus corresponding baseline row.

sentiment and position recognition) persist regardless of the temperature settings. This stability across different generation hyperparameters further validates the reliability of our multi-granularity evaluation protocol. The detailed results for temperatures 0, 0.7, and 1.0 are reported in Tables 2, 3, and 4, respectively.

5 Additional Multi-Granularity Evaluation Results

Figure 2 illustrates the complete multi-granularity accuracy curves for the remaining nine LVLMs evaluated on LIVE under Uniform Image Context (UIC) and Diverse Image Context (DIC) settings at a temperature of 0. Additionally, the results for all models at temperatures of 0.7 and 1.0 are provided in Figure 3 and Figure 4, respectively.

6 Human Validation and API Expenses

To ensure the correctness and high quality of our dataset, we primarily rely on Anchor-based Controlled Generation, which structurally precludes accidental correctness by enforcing context-conflict insertions. Because of this, the generated negatives are inherently incompatible with the image content. However, to guarantee absolute rigor, human verification serves as our final safeguard.

Human Validation Protocol. We recruited 4 expert annotators (graduate researchers with computer vision backgrounds) to perform double-blind validation. Annotators were instructed to verify whether the LLM-generated counterfactual items were strictly absent from all relevant images while remaining contextually plausible within the broader scenario. The annotators achieved a 98.5% label agreement and a 0.92 Cohen's Kappa score. During this filtering process, approximately 4.2% of the initial generated samples were rejected or revised. This low rejection rate demonstrates both the high initial accuracy of our automated pipeline and the strictness of our quality control.

API and Computational Expenses. Our semi-automated pipeline is highly scalable and cost-efficient. The scene graph parsing step is computationally negligible. For the LLM-assisted counterfactual generation, rather than relying on expensive closed-source APIs, we deployed the open-weight Qwen2.5-72B-Instruct model locally. This required only ∼4 hours of inference time on 2 × H200 GPUs. Consequently, direct API expenses were zero, and the primary pipeline costs were strictly limited to accessible GPU compute hours and human verification compensation.

7 CLIP Similarity Gap

To quantitatively validate the distinction between Uniform Image Contexts (UIC) and Diverse Image Contexts (DIC), we calculated the average intra-group semantic similarity for samples consisting of four images.

As shown in Table 5, we calculated the intra-group CLIP ViT-B/32 cosine similarity across all 6 tasks. The significant gap ($\Delta \approx 0.1$) confirms our scenarios possess distinct semantic distributions: UIC implies high visual cohesion (consistent lighting, backgrounds, and themes, e.g., Pizza in Fig. 2), whereas DIC spans broad, heterogeneous contexts.

8 Analysis of "Any" vs. "Each" Logical Forms

As noted in our main paper, the LIVE benchmark includes two distinct logical formulations for multi-image queries: existential quantification (*"Is [⋆] in **any** of..."*) and universal quantification (*"Is [⋆] in **each** of..."*). These impose fundamentally different reasoning demands. The "any" condition requires the model to correctly identify at least one positive occurrence across multiple images (where a single presence yields a Yes), whereas the "each" condition

Table 2: Full results of performance with MAP across different targeted images at temperature = 0.

Task		GPT-4o		Qwen3-32B		InternVL-26B		Qwen3-8B		InternVL-8B		MiniCPM-4.5		MiniCPM-2.6		Mantis-8B		Qwen2.5-7B		LLaVA-OV		LLaVA-IL	
Class	**TI**	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc
Scenario 1: Uniform Image Context																							
Object	TI_1	0.969	0.969	0.963	0.964	0.896	0.904	**0.969**	**0.969**	0.815	0.845	0.961	0.962	0.959	0.960	0.894	0.890	0.892	0.903	0.925	0.926	0.901	0.896
	TI_2	0.977	0.977	**0.954**	**0.956**	0.910	0.916	0.902	0.911	0.864	0.880	0.931	0.935	0.850	0.870	0.837	0.856	0.772	0.814	0.916	0.922	0.951	0.950
	TI_3	0.979	0.979	0.958	0.960	0.915	0.921	0.925	0.930	0.875	0.888	0.942	0.945	0.794	0.829	0.850	0.869	0.741	0.795	0.926	0.931	**0.962**	**0.961**
	TI_4	0.971	0.971	**0.978**	**0.978**	0.925	0.931	0.950	0.953	0.904	0.912	0.935	0.938	0.835	0.858	0.921	0.927	0.762	0.807	0.931	0.934	0.968	0.967
	Avg	0.974	0.974	**0.963**	**0.964**	0.912	0.918	0.936	0.941	0.864	0.881	0.942	0.945	0.860	0.879	0.876	0.885	0.792	0.830	0.924	0.928	0.945	0.944
Material	TI_1	0.952	0.950	**0.934**	**0.940**	0.873	0.898	0.924	0.934	0.848	0.875	0.922	0.926	0.902	0.910	0.917	0.914	0.849	0.871	0.913	0.923	0.893	0.884
	TI_2	0.898	0.902	0.852	0.871	0.855	0.864	0.811	0.833	0.763	0.802	0.835	0.855	0.668	0.750	**0.885**	0.884	0.393	0.623	**0.885**	**0.893**	0.869	0.852
	TI_3	0.927	0.931	0.890	0.899	0.824	0.845	0.861	0.873	0.857	0.873	0.872	0.882	0.634	0.731	0.865	0.869	0.285	0.583	**0.901**	**0.907**	0.889	0.881
	TI_4	0.895	0.900	**0.912**	**0.915**	0.865	0.875	0.902	0.905	0.871	0.880	0.842	0.855	0.713	0.775	0.857	0.860	0.261	0.575	0.883	0.890	0.893	0.885
	Avg	0.918	0.921	**0.897**	**0.906**	0.854	0.870	0.874	0.886	0.835	0.857	0.868	0.880	0.729	0.791	0.881	0.882	0.447	0.663	0.896	0.903	0.886	0.875
Color	TI_1	0.975	0.975	**0.957**	**0.959**	0.854	0.871	0.937	0.941	0.800	0.832	0.936	0.940	0.933	0.936	0.853	0.844	0.871	0.889	0.852	0.870	0.895	0.891
	TI_2	0.913	0.918	0.872	0.886	**0.890**	**0.896**	0.821	0.849	0.844	0.860	0.813	0.840	0.624	0.728	0.870	0.867	0.450	0.644	0.837	0.856	0.874	0.859
	TI_3	0.908	0.914	**0.912**	**0.918**	0.838	0.859	0.859	0.877	0.883	0.893	0.887	0.896	0.532	0.681	0.866	0.869	0.346	0.605	0.894	0.902	0.903	0.895
	TI_4	0.941	0.942	**0.960**	**0.960**	0.879	0.890	0.947	0.948	0.907	0.913	0.886	0.896	0.735	0.790	0.890	0.892	0.348	0.605	0.905	0.912	0.922	0.918
	Avg	0.934	0.937	**0.925**	**0.931**	0.865	0.879	0.891	0.904	0.859	0.875	0.881	0.893	0.706	0.784	0.870	0.868	0.503	0.686	0.872	0.885	0.899	0.891
Sentiment	TI_1	0.925	0.930	0.930	0.935	0.883	0.895	0.863	0.887	0.849	0.872	**0.968**	**0.969**	0.876	0.880	0.805	0.768	0.857	0.873	0.858	0.866	0.876	0.869
	TI_2	0.835	0.856	**0.916**	**0.924**	**0.916**	0.914	0.829	0.869	0.834	0.851	0.811	0.829	0.702	0.764	0.850	0.841	0.553	0.696	0.883	0.890	0.869	0.848
	TI_3	0.751	0.796	0.948	0.951	0.902	0.910	0.905	0.917	**0.952**	**0.954**	0.874	0.879	0.721	0.783	0.897	0.896	0.651	0.749	0.938	0.942	0.914	0.905
	TI_4	0.866	0.882	**0.974**	**0.974**	0.904	0.908	0.959	0.961	0.960	0.961	0.917	0.921	0.882	0.895	0.909	0.908	0.712	0.776	0.973	**0.974**	0.927	0.921
	Avg	0.844	0.866	**0.942**	**0.946**	0.901	0.907	0.889	0.908	0.899	0.909	0.893	0.900	0.795	0.830	0.865	0.853	0.693	0.774	0.913	0.918	0.896	0.886
Action	TI_1	0.923	0.927	0.902	0.910	0.869	0.880	0.867	0.882	0.752	0.803	**0.936**	**0.938**	0.884	0.892	0.851	0.840	0.780	0.819	0.901	0.906	0.887	0.884
	TI_2	0.746	0.796	0.730	0.788	0.863	0.875	0.660	0.750	0.692	0.763	0.759	0.802	0.503	0.668	0.759	0.785	0.344	0.604	0.802	0.831	**0.884**	**0.882**
	TI_3	0.780	0.818	0.798	0.831	0.814	0.843	0.696	0.766	0.782	0.819	0.835	0.854	0.386	0.618	0.800	0.820	0.285	0.583	0.805	0.833	**0.882**	**0.883**
	TI_4	0.884	0.896	0.869	0.880	0.826	0.844	0.812	0.839	0.817	0.844	0.864	0.879	0.638	0.734	0.795	0.823	0.286	0.583	0.874	0.885	**0.895**	**0.896**
	Avg	0.833	0.859	0.825	0.852	0.843	0.861	0.759	0.809	0.761	0.807	0.848	0.868	0.603	0.728	0.801	0.817	0.424	0.647	0.845	0.864	**0.887**	**0.886**
Position	TI_1	0.925	0.926	0.922	0.927	0.845	0.864	0.892	0.902	0.744	0.798	**0.940**	**0.941**	0.853	0.866	0.835	0.816	0.730	0.790	0.856	0.865	0.855	0.840
	TI_2	0.668	0.749	0.729	0.787	0.853	**0.865**	0.675	0.757	0.667	0.748	0.717	0.774	0.399	0.622	0.736	0.765	0.290	0.586	0.719	0.776	**0.854**	0.836
	TI_3	0.712	0.775	0.762	0.806	0.794	0.825	0.694	0.765	0.765	0.807	0.748	0.794	0.357	0.608	0.770	0.789	0.277	0.582	0.785	0.819	**0.874**	**0.866**
	TI_4	0.780	0.820	0.870	**0.885**	0.781	0.815	0.832	0.855	0.826	0.850	0.839	0.859	0.582	0.705	0.832	0.840	0.361	0.610	0.798	0.825	**0.887**	**0.885**
	Avg	0.771	0.818	0.821	0.851	0.818	0.842	0.774	0.820	0.750	0.801	0.811	0.842	0.547	0.700	0.793	0.803	0.414	0.642	0.789	0.821	**0.867**	**0.857**
Scenario 2: Diverse Image Context																							
Object	TI_1	0.980	0.979	**0.979**	**0.979**	0.875	0.890	**0.979**	**0.979**	0.790	0.830	**0.979**	**0.979**	0.972	0.973	0.867	0.866	0.909	0.917	0.913	0.917	0.900	0.899
	TI_2	0.971	0.971	**0.960**	**0.961**	0.795	0.829	0.877	0.891	0.678	0.759	0.878	0.891	0.696	0.767	0.740	0.785	0.543	0.688	0.774	0.812	0.932	0.928
	TI_3	0.975	0.975	**0.961**	**0.962**	0.802	0.834	0.856	0.876	0.758	0.806	0.872	0.887	0.522	0.676	0.837	0.856	0.442	0.643	0.818	0.845	0.937	0.935
	TI_4	0.957	0.957	**0.952**	**0.952**	0.786	0.817	0.902	0.909	0.813	0.838	0.845	0.861	0.658	0.745	0.793	0.817	0.375	0.615	0.842	0.861	0.916	0.913
	Avg	0.971	0.971	**0.963**	**0.963**	0.815	0.843	0.904	0.914	0.760	0.808	0.894	0.905	0.712	0.790	0.809	0.831	0.567	0.716	0.837	0.859	0.921	0.919
Material	TI_1	0.957	0.959	0.945	0.946	0.907	0.911	0.942	0.944	0.803	0.839	**0.950**	**0.953**	0.910	0.915	0.861	0.851	0.868	0.886	0.897	0.898	0.862	0.844
	TI_2	0.904	0.912	0.797	0.837	0.838	0.852	0.749	0.799	0.685	0.759	0.724	0.784	0.502	0.668	0.814	0.820	0.346	0.606	0.845	**0.860**	**0.852**	0.844
	TI_3	0.919	0.923	0.864	0.880	0.790	0.822	0.785	0.824	0.726	0.785	0.824	0.852	0.379	0.617	0.827	0.835	0.214	0.560	**0.874**	**0.885**	0.851	0.854
	TI_4	0.941	0.944	**0.910**	**0.916**	0.800	0.827	0.876	0.888	0.749	0.799	0.826	0.851	0.560	0.692	0.847	0.855	0.171	0.547	0.901	0.906	0.856	0.855
	Avg	0.930	0.934	**0.879**	**0.895**	0.834	0.853	0.838	0.864	0.741	0.796	0.831	0.860	0.588	0.723	0.837	0.840	0.400	0.650	**0.879**	0.887	0.855	0.849
Color	TI_1	0.967	0.968	**0.959**	**0.961**	0.905	0.914	0.943	0.946	0.897	0.907	0.949	0.951	0.928	0.932	0.876	0.871	0.906	0.917	0.926	0.931	0.935	0.933
	TI_2	0.909	0.916	0.864	0.880	0.843	0.862	0.701	0.770	0.820	0.849	0.690	0.762	0.442	0.644	0.840	0.843	0.228	0.566	0.845	0.866	**0.894**	**0.884**
	TI_3	0.902	0.909	0.902	**0.910**	0.684	0.760	0.761	0.807	0.828	0.852	0.822	0.847	0.312	0.593	0.824	0.834	0.106	0.528	0.889	0.899	**0.907**	0.901
	TI_4	0.941	0.944	**0.950**	**0.951**	0.728	0.783	0.887	0.895	0.819	0.845	0.793	0.826	0.567	0.696	0.844	0.851	0.116	0.531	0.914	0.919	0.911	0.905
	Avg	0.930	0.934	**0.919**	**0.925**	0.790	0.830	0.823	0.855	0.841	0.863	0.814	0.846	0.562	0.717	0.846	0.850	0.339	0.635	0.893	0.904	0.912	0.906
Sentiment	TI_1	0.928	0.929	0.811	0.827	0.794	0.833	0.813	0.845	0.733	0.788	0.837	0.848	**0.893**	**0.894**	0.840	0.814	0.769	0.803	0.850	0.848	0.827	0.818
	TI_2	0.747	0.800	**0.845**	**0.860**	0.743	0.771	0.626	0.717	0.761	0.801	0.731	0.763	0.552	0.696	0.844	0.856	0.183	0.530	0.792	0.814	**0.845**	0.835
	TI_3	0.576	0.699	**0.880**	**0.883**	0.812	0.824	0.758	0.791	0.784	0.807	0.807	0.813	0.471	0.649	0.782	0.810	0.244	0.574	0.806	0.815	0.873	0.870
	TI_4	0.696	0.767	0.881	0.883	0.877	0.883	0.836	0.850	0.857	0.867	0.764	0.783	0.857	0.867	0.792	0.817	0.500	0.667	0.881	0.883	**0.903**	**0.900**
	Avg	0.737	0.799	0.854	**0.864**	0.807	0.828	0.758	0.801	0.784	0.816	0.785	0.802	0.693	0.776	0.815	0.824	0.424	0.644	0.832	0.840	**0.862**	0.856
Action	TI_1	0.948	0.950	0.919	0.924	0.892	0.901	0.880	0.892	0.810	0.841	**0.938**	**0.939**	0.891	0.898	0.850	0.845	0.816	0.845	0.900	0.905	0.893	0.895
	TI_2	0.639	0.732	0.690	0.766	0.745	0.796	0.437	0.645	0.529	0.679	0.646	0.735	0.207	0.558	0.650	0.718	0.087	0.523	0.616	0.722	**0.864**	**0.871**
	TI_3	0.730	0.789	0.758	0.811	0.643	0.736	0.614	0.725	0.659	0.746	0.819	0.846	0.129	0.536	0.698	0.750	0.069	0.518	0.668	0.747	**0.872**	**0.878**
	TI_4	0.828	0.854	0.851	0.869	0.680	0.758	0.747	0.798	0.717	0.773	0.814	0.843	0.390	0.621	0.695	0.742	0.059	0.515	0.750	0.798	**0.868**	**0.874**
	Avg	0.786	0.831	0.805	0.842	0.740	0.798	0.669	0.765	0.679	0.760	0.804	0.841	0.404	0.653	0.723	0.764	0.258	0.600	0.734	0.793	**0.874**	**0.879**
Position	TI_1	0.939	0.940	**0.929**	**0.933**	0.852	0.871	0.882	0.894	0.765	0.806	**0.929**	0.930	0.853	0.866	0.823	0.806	0.746	0.799	0.870	0.876	0.846	0.834
	TI_2	0.579	0.699	0.587	0.707	0.743	0.785	0.468	0.657	0.478	0.655	0.522	0.671	0.161	0.542	0.623	0.699	0.067	0.517	0.549	0.682	**0.855**	**0.850**
	TI_3	0.673	0.752	0.634	0.733	0.590	0.703	0.497	0.666	0.545	0.679	0.662	0.746	0.131	0.535	0.687	0.736	0.058	0.514	0.586	0.703	**0.804**	**0.814**
	TI_4	0.722	0.780	0.821	**0.845**	0.667	0.740	0.725	0.780	0.701	0.765	0.751	0.795	0.333	0.600	0.756	0.780	0.077	0.520	0.739	0.785	**0.833**	0.830
	Avg	0.728	0.793	0.743	0.804	0.713	0.775	0.643	0.749	0.622	0.726	0.716	0.786	0.370	0.636	0.722	0.755	0.237	0.588	0.686	0.761	**0.834**	**0.832**

Note: **Bold** and underlined denote best and second-best results (excluding GPT-4o).

requires the model to verify presence in all queried images (where a single absence is sufficient to yield a No).

To investigate whether models exhibit varying sensitivities to these logical structures, we provide a detailed breakdown of accuracy under both formulations across different sampling temperatures ($t = 0, 0.7, 1.0$). As shown in Table 6, a consistent performance

Table 3: Full results of performance with MAP across different targeted images at temperature = 0.7.

Task		GPT-4o		Qwen3-32B		InternVL-26B		Qwen3-8B		InternVL-8B		MiniCPM-4.5		MiniCPM-2.6		Mantis-8B		Qwen2.5-7B		LLaVA-OV		LLaVA-IL	
Class	**TI**	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc
Scenario 1: Uniform Image Context																							
Object	TI_1	0.968	0.968	0.964	0.964	0.866	0.877	**0.969**	**0.969**	0.786	0.819	0.959	0.960	0.944	0.945	0.859	0.856	0.864	0.879	0.822	0.834	0.861	0.853
	TI_2	0.979	0.979	**0.955**	**0.957**	0.880	0.890	0.907	0.915	0.824	0.849	0.920	0.925	0.844	0.863	0.785	0.813	0.755	0.803	0.816	0.837	0.928	0.925
	TI_3	0.981	0.981	**0.958**	**0.960**	0.901	0.909	0.926	0.931	0.843	0.864	0.923	0.928	0.811	0.840	0.820	0.842	0.735	0.791	0.833	0.852	0.932	0.930
	TI_4	0.975	0.975	**0.978**	**0.978**	0.918	0.923	0.962	0.964	0.896	0.905	0.927	0.931	0.865	0.880	0.862	0.876	0.743	0.796	0.877	0.887	0.935	0.934
	Avg	0.976	0.975	**0.964**	**0.965**	0.892	0.900	0.941	0.945	0.838	0.859	0.932	0.936	0.866	0.882	0.832	0.847	0.774	0.817	0.837	0.852	0.914	0.911
Material	TI_1	0.948	0.945	**0.939**	**0.945**	0.838	0.859	0.927	0.936	0.800	0.833	0.916	0.918	0.885	0.891	0.842	0.834	0.806	0.837	0.819	0.824	0.855	0.838
	TI_2	0.900	0.904	**0.850**	**0.869**	0.801	0.814	0.812	0.833	0.747	0.783	0.815	0.838	0.668	0.747	0.796	0.802	0.470	0.651	0.750	0.780	0.815	0.791
	TI_3	0.919	0.923	**0.892**	**0.901**	0.803	0.822	0.877	0.886	0.805	0.830	0.870	0.879	0.672	0.742	0.783	0.789	0.343	0.603	0.811	0.828	0.833	0.819
	TI_4	0.912	0.915	**0.912**	**0.915**	0.816	0.835	0.893	0.895	0.811	0.830	0.802	0.825	0.649	0.730	0.755	0.770	0.333	0.600	0.808	0.820	0.852	0.835
	Avg	0.920	0.922	**0.898**	**0.907**	0.814	0.833	0.877	0.887	0.791	0.819	0.851	0.865	0.718	0.778	0.794	0.799	0.488	0.673	0.797	0.813	0.839	0.821
Color	TI_1	0.974	0.974	**0.957**	**0.959**	0.822	0.845	0.935	0.939	0.775	0.812	0.941	0.944	0.933	0.935	0.819	0.809	0.855	0.873	0.791	0.816	0.863	0.856
	TI_2	0.913	0.917	**0.870**	**0.884**	0.843	0.853	0.818	0.848	0.814	0.830	0.803	0.832	0.645	0.737	0.781	0.779	0.471	0.653	0.758	0.791	0.825	0.800
	TI_3	0.906	0.912	**0.908**	**0.914**	0.808	0.833	0.865	0.881	0.849	0.862	0.869	0.881	0.580	0.702	0.797	0.800	0.387	0.620	0.811	0.832	0.843	0.829
	TI_4	0.942	0.944	**0.963**	**0.963**	0.848	0.861	0.941	0.942	0.863	0.872	0.866	0.880	0.725	0.779	0.813	0.816	0.396	0.624	0.855	0.869	0.873	0.861
	Avg	0.934	0.937	**0.924**	**0.930**	0.830	0.848	0.890	0.902	0.825	0.844	0.870	0.884	0.721	0.788	0.803	0.801	0.527	0.692	0.804	0.827	0.851	0.837
Sentiment	TI_1	0.918	0.924	**0.930**	**0.935**	0.873	0.877	0.852	0.877	0.787	0.822	0.890	0.901	0.851	0.859	0.813	0.788	0.834	0.850	0.815	0.834	0.802	0.779
	TI_2	0.855	0.870	**0.924**	**0.928**	0.884	0.885	0.829	0.869	0.790	0.813	0.818	0.835	0.720	0.773	0.816	0.798	0.524	0.674	0.789	0.806	0.844	0.826
	TI_3	0.734	0.783	**0.954**	**0.956**	0.897	0.902	0.925	0.934	0.897	0.901	0.833	0.836	0.713	0.776	0.803	0.800	0.625	0.732	0.877	0.884	0.844	0.824
	TI_4	0.882	0.895	**0.962**	**0.961**	0.849	0.855	0.959	**0.961**	0.917	0.921	0.917	0.921	0.822	0.829	0.821	0.816	0.667	0.750	0.841	0.855	0.819	0.803
	Avg	0.847	0.868	**0.943**	**0.945**	0.876	0.880	0.891	0.910	0.848	0.864	0.864	0.873	0.777	0.809	0.813	0.800	0.662	0.751	0.831	0.845	0.828	0.808
Action	TI_1	0.932	0.934	0.905	0.912	0.824	0.844	0.860	0.877	0.738	0.785	**0.925**	**0.927**	0.858	0.866	0.792	0.779	0.771	0.811	0.805	0.820	0.848	0.843
	TI_2	0.762	0.806	0.735	0.791	0.823	**0.840**	0.671	0.755	0.689	0.751	0.750	0.790	0.572	0.690	0.718	0.742	0.370	0.612	0.713	0.762	**0.840**	0.834
	TI_3	0.764	0.806	0.802	0.833	0.785	0.816	0.689	0.762	0.740	0.786	**0.830**	**0.849**	0.442	0.637	0.728	0.755	0.301	0.587	0.769	0.799	0.812	0.810
	TI_4	0.871	0.885	**0.881**	**0.891**	0.800	0.823	0.812	0.839	0.812	0.839	0.802	0.828	0.600	0.708	0.726	0.760	0.397	0.620	0.732	0.771	0.863	0.865
	Avg	0.832	0.858	0.831	**0.857**	0.808	0.831	0.758	0.808	0.745	0.790	0.827	0.849	0.618	0.725	0.741	0.759	0.460	0.658	0.755	0.788	**0.841**	0.838
Position	TI_1	0.922	0.923	0.922	**0.927**	0.804	0.826	0.892	0.902	0.738	0.782	**0.925**	0.926	0.837	0.850	0.768	0.751	0.721	0.782	0.782	0.797	0.818	0.802
	TI_2	0.690	0.761	0.726	0.786	**0.818**	**0.831**	0.686	0.762	0.681	0.743	0.719	0.771	0.471	0.647	0.690	0.713	0.328	0.599	0.684	0.736	0.817	0.800
	TI_3	0.696	0.765	0.770	**0.812**	0.757	0.792	0.699	0.767	0.757	0.792	0.756	0.795	0.441	0.636	0.735	0.746	0.299	0.588	0.721	0.759	**0.807**	0.798
	TI_4	0.788	0.825	**0.864**	**0.880**	0.765	0.800	0.826	0.850	0.790	0.825	0.823	0.845	0.547	0.685	0.735	0.755	0.387	0.620	0.775	0.805	0.847	0.845
	Avg	0.774	0.818	0.821	**0.851**	0.786	0.812	0.776	0.820	0.742	0.786	0.806	0.834	0.574	0.704	0.732	0.741	0.434	0.647	0.740	0.774	**0.823**	0.811
Scenario 2: Diverse Image Context																							
Object	TI_1	0.979	0.979	0.977	**0.978**	0.851	0.871	**0.978**	**0.978**	0.777	0.812	0.965	0.965	0.959	0.960	0.823	0.824	0.894	0.904	0.820	0.834	0.875	0.873
	TI_2	0.967	0.967	**0.961**	**0.963**	0.788	0.820	0.877	0.890	0.667	0.746	0.868	0.881	0.694	0.765	0.734	0.772	0.553	0.692	0.719	0.766	0.909	0.903
	TI_3	0.975	0.975	**0.962**	**0.963**	0.781	0.816	0.861	0.878	0.744	0.796	0.867	0.883	0.552	0.691	0.777	0.808	0.479	0.658	0.759	0.799	0.919	0.917
	TI_4	0.967	0.966	**0.948**	**0.947**	0.777	0.812	0.885	0.894	0.856	0.870	0.888	0.894	0.688	0.760	0.769	0.798	0.375	0.615	0.718	0.769	0.910	0.909
	Avg	0.972	0.972	**0.962**	**0.962**	0.799	0.830	0.900	0.910	0.761	0.806	0.897	0.906	0.723	0.794	0.776	0.800	0.575	0.717	0.754	0.792	0.903	0.900
Material	TI_1	0.948	0.949	**0.940**	**0.943**	0.881	0.889	0.933	0.936	0.795	0.826	0.917	0.922	0.895	0.901	0.775	0.756	0.848	0.867	0.830	0.834	0.835	0.825
	TI_2	0.917	0.922	**0.798**	**0.838**	0.789	0.805	0.763	0.809	0.680	0.749	0.751	0.797	0.554	0.686	0.745	0.748	0.430	0.639	0.765	0.791	0.773	0.761
	TI_3	0.911	0.917	**0.862**	**0.879**	0.750	0.787	0.790	0.829	0.735	0.785	0.790	0.824	0.450	0.641	0.770	0.772	0.269	0.579	0.800	0.820	0.792	0.786
	TI_4	0.956	0.958	**0.904**	**0.911**	0.722	0.766	0.860	0.874	0.746	0.790	0.800	0.827	0.566	0.692	0.750	0.757	0.246	0.570	0.833	0.846	0.778	0.771
	Avg	0.933	0.936	**0.876**	**0.893**	0.786	0.812	0.837	0.862	0.739	0.787	0.814	0.842	0.616	0.730	0.760	0.758	0.448	0.664	0.807	0.823	0.795	0.786
Color	TI_1	0.967	0.968	**0.960**	**0.961**	0.879	0.890	0.947	0.949	0.868	0.883	0.943	0.945	0.925	0.929	0.809	0.797	0.891	0.903	0.860	0.872	0.893	0.888
	TI_2	0.908	0.915	**0.868**	**0.883**	0.809	0.831	0.703	0.772	0.768	0.804	0.706	0.770	0.524	0.677	0.735	0.738	0.313	0.593	0.754	0.792	0.837	0.819
	TI_3	0.901	0.909	**0.903**	**0.911**	0.668	0.746	0.765	0.810	0.795	0.824	0.803	0.832	0.399	0.623	0.741	0.748	0.197	0.555	0.808	0.830	0.834	0.823
	TI_4	0.938	0.941	**0.943**	**0.945**	0.702	0.764	0.893	0.900	0.787	0.818	0.794	0.826	0.622	0.724	0.740	0.747	0.169	0.546	0.818	0.837	0.842	0.826
	Avg	0.929	0.933	**0.919**	**0.925**	0.765	0.808	0.827	0.858	0.804	0.832	0.811	0.843	0.618	0.738	0.756	0.758	0.392	0.649	0.810	0.833	0.852	0.839
Sentiment	TI_1	0.910	0.909	0.811	0.827	0.742	0.788	0.813	0.845	0.784	0.809	0.822	0.836	**0.884**	**0.884**	0.815	0.776	0.770	0.806	0.740	0.774	0.783	0.756
	TI_2	0.747	0.796	**0.854**	**0.866**	0.757	0.794	0.651	0.732	0.724	0.756	0.763	0.796	0.562	0.687	0.667	0.665	0.246	0.552	0.730	0.762	0.779	0.760
	TI_3	0.593	0.706	**0.880**	**0.883**	0.808	0.827	0.765	0.797	0.777	0.802	0.784	0.790	0.546	0.673	0.706	0.737	0.287	0.566	0.766	0.791	0.790	0.775
	TI_4	0.723	0.783	**0.881**	**0.883**	0.857	0.867	0.836	0.850	0.815	0.833	0.828	0.833	0.848	0.850	0.737	0.750	0.500	0.667	0.793	0.800	0.831	0.817
	Avg	0.743	0.799	**0.857**	**0.865**	0.791	0.819	0.766	0.806	0.775	0.800	0.799	0.814	0.710	0.773	0.731	0.732	0.451	0.648	0.757	0.782	0.796	0.777
Action	TI_1	0.945	0.947	0.919	0.923	0.842	0.858	0.878	0.890	0.790	0.823	**0.933**	**0.934**	0.871	0.878	0.793	0.789	0.768	0.811	0.826	0.837	0.847	0.848
	TI_2	0.642	0.735	0.696	0.770	0.730	0.780	0.465	0.656	0.576	0.695	0.637	0.729	0.306	0.586	0.625	0.695	0.137	0.537	0.604	0.704	**0.794**	**0.801**
	TI_3	0.750	0.799	0.761	0.812	0.635	0.727	0.621	0.728	0.629	0.723	0.817	0.841	0.221	0.560	0.682	0.721	0.078	0.520	0.662	0.735	**0.841**	**0.849**
	TI_4	0.807	0.838	**0.846**	**0.864**	0.658	0.727	0.755	0.803	0.679	0.742	0.814	0.843	0.470	0.646	0.757	0.783	0.096	0.525	0.654	0.722	0.806	0.818
	Avg	0.786	0.830	0.805	**0.842**	0.716	0.773	0.679	0.769	0.668	0.746	0.800	0.837	0.467	0.668	0.714	0.747	0.270	0.598	0.686	0.749	**0.822**	0.829
Position	TI_1	0.931	0.933	**0.934**	**0.937**	0.807	0.828	0.888	0.899	0.747	0.786	0.918	0.919	0.823	0.838	0.746	0.731	0.731	0.785	0.798	0.810	0.824	0.811
	TI_2	0.575	0.698	0.592	0.710	0.706	0.754	0.470	0.658	0.503	0.652	0.543	0.677	0.289	0.578	0.633	0.684	0.088	0.522	0.547	0.660	**0.791**	**0.788**
	TI_3	0.670	0.751	0.637	0.734	0.605	0.706	0.509	0.671	0.574	0.685	0.680	0.756	0.195	0.549	0.660	0.703	0.090	0.521	0.611	0.701	**0.751**	**0.760**
	TI_4	0.768	0.810	**0.826**	**0.850**	0.667	0.740	0.721	0.775	0.683	0.748	0.765	0.800	0.361	0.610	0.637	0.670	0.113	0.530	0.725	0.780	0.769	0.775
	Avg	0.736	0.798	0.747	**0.808**	0.696	0.757	0.647	0.751	0.627	0.718	0.726	0.788	0.417	0.644	0.669	0.697	0.256	0.590	0.670	0.738	**0.784**	0.784

Note: **Bold** and underlined denote best and second-best results (excluding GPT-4o).

gap exists across all tested LVLMs, where accuracy on "each" queries is systematically higher than on "any" queries. Several key insights emerge from this breakdown:

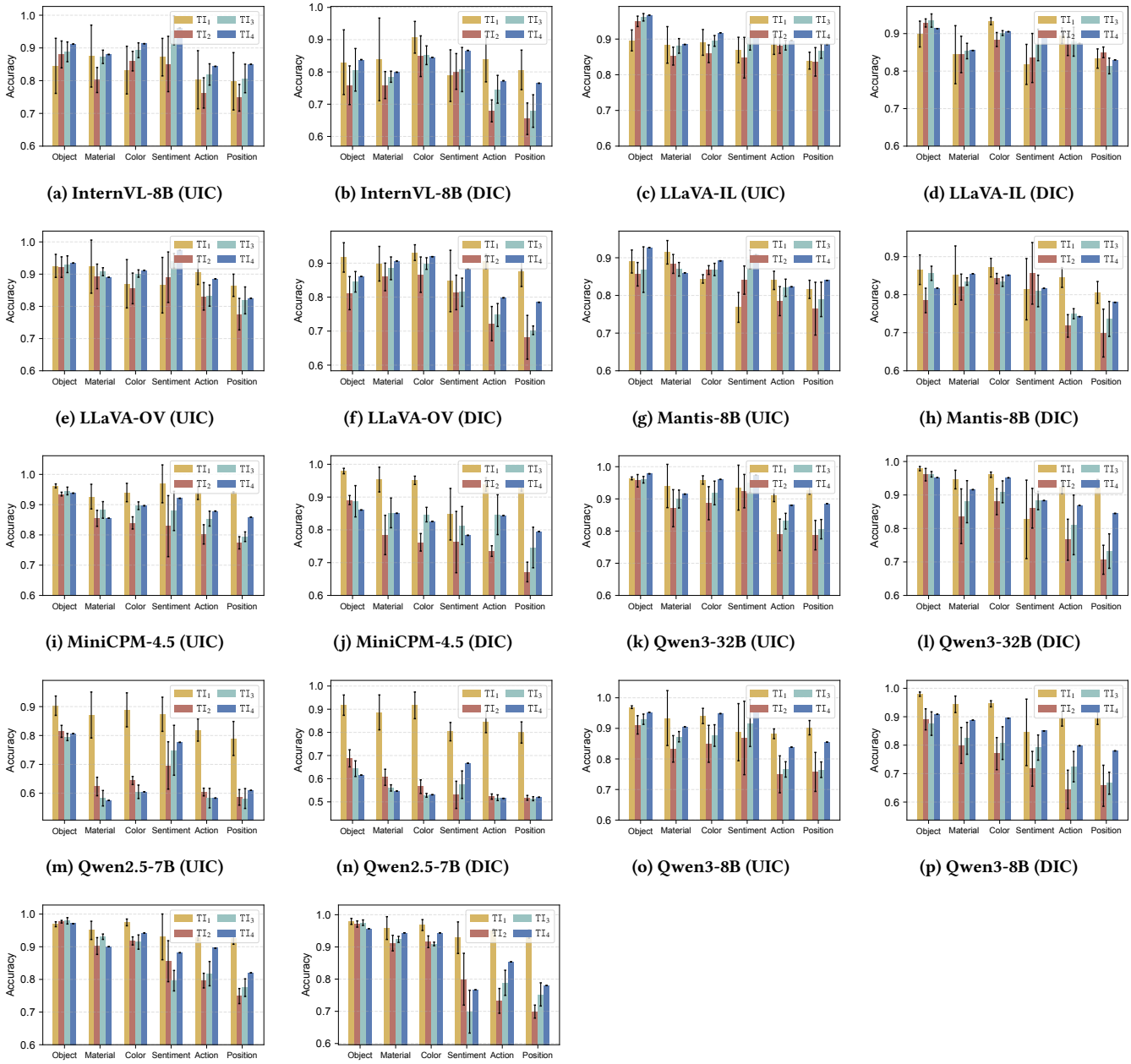
❶ **Struggle with Existential Reasoning.** Models are generally more prone to errors when evaluating "any" conditions. This suggests that LVLMs might struggle to confidently aggregate positive

Table 4: Full results of performance with MAP across different targeted images at temperature = 1.0.

Task		GPT-4o		Qwen3-32B		InternVL-26B		Qwen3-8B		InternVL-8B		MiniCPM-4.5		MiniCPM-2.6		Mantis-8B		Qwen2.5-7B		LLaVA-OV		LLaVA-IL	
Class	**TI**	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc
Scenario 1: Uniform Image Context																							
Object	TI_1	0.969	0.969	0.962	0.963	0.853	0.864	**0.970**	**0.970**	0.763	0.802	0.945	0.946	0.931	0.932	0.797	0.796	0.845	0.863	0.781	0.794	0.831	0.824
	TI_2	0.974	0.974	**0.954**	**0.956**	0.876	0.887	0.908	0.916	0.806	0.832	0.911	0.917	0.828	0.851	0.740	0.770	0.750	0.800	0.774	0.802	0.908	0.904
	TI_3	0.984	0.984	**0.962**	**0.963**	0.877	0.889	0.931	0.936	0.823	0.843	0.902	0.909	0.786	0.823	0.771	0.797	0.749	0.799	0.804	0.829	0.911	0.908
	TI_4	0.975	0.975	**0.978**	**0.978**	0.909	0.916	0.962	0.964	0.855	0.872	0.931	0.934	0.831	0.854	0.800	0.825	0.757	0.803	0.787	0.814	0.885	0.880
	Avg	0.976	0.975	**0.964**	**0.965**	0.879	0.889	0.943	0.946	0.812	0.838	0.922	0.927	0.844	0.865	0.777	0.797	0.775	0.816	0.786	0.810	0.884	0.879
Material	TI_1	0.952	0.950	0.931	0.938	0.805	0.834	**0.934**	**0.941**	0.766	0.806	0.904	0.909	0.871	0.873	0.794	0.785	0.822	0.847	0.770	0.788	0.830	0.815
	TI_2	0.896	0.900	**0.848**	**0.868**	0.776	0.792	0.840	0.855	0.692	0.737	0.805	0.829	0.678	0.744	0.707	0.717	0.503	0.667	0.737	0.761	0.784	0.761
	TI_3	0.924	0.928	**0.883**	**0.893**	0.800	0.818	0.866	0.876	0.779	0.803	0.840	0.851	0.635	0.718	0.738	0.745	0.352	0.604	0.750	0.779	0.803	0.790
	TI_4	0.902	0.905	**0.923**	**0.925**	0.753	0.780	0.898	0.900	0.735	0.765	0.829	0.845	0.726	0.770	0.751	0.775	0.481	0.655	0.757	0.775	0.779	0.765
	Avg	0.918	0.921	**0.896**	**0.906**	0.784	0.806	0.885	0.893	0.743	0.778	0.844	0.858	0.728	0.776	0.748	0.756	0.540	0.693	0.753	0.776	0.799	0.783
Color	TI_1	0.975	0.975	**0.958**	**0.959**	0.824	0.846	0.935	0.938	0.777	0.804	0.928	0.932	0.910	0.914	0.752	0.739	0.837	0.859	0.747	0.776	0.804	0.794
	TI_2	0.915	0.920	**0.872**	**0.886**	0.809	0.822	0.820	0.849	0.777	0.800	0.785	0.817	0.644	0.730	0.733	0.731	0.525	0.675	0.732	0.769	0.783	0.756
	TI_3	0.910	0.915	**0.910**	**0.916**	0.781	0.807	0.863	0.879	0.812	0.829	0.855	0.869	0.572	0.693	0.732	0.738	0.403	0.625	0.757	0.785	0.803	0.785
	TI_4	0.937	0.939	**0.960**	**0.960**	0.793	0.817	0.946	0.947	0.820	0.834	0.842	0.860	0.732	0.779	0.788	0.788	0.394	0.620	0.783	0.808	0.844	0.828
	Avg	0.934	0.937	**0.925**	**0.931**	0.802	0.823	0.891	0.903	0.796	0.817	0.853	0.870	0.715	0.779	0.751	0.749	0.540	0.695	0.755	0.785	0.809	0.791
Sentiment	TI_1	0.923	0.928	**0.930**	**0.935**	0.858	0.871	0.838	0.867	0.753	0.778	0.922	0.923	0.796	0.797	0.728	0.677	0.811	0.832	0.745	0.754	0.741	0.725
	TI_2	0.844	0.862	**0.910**	**0.918**	0.888	0.886	0.841	0.878	0.751	0.771	0.805	0.812	0.696	0.746	0.738	0.727	0.566	0.690	0.759	0.764	0.837	0.811
	TI_3	0.772	0.810	**0.940**	**0.942**	0.798	0.812	0.901	0.916	0.847	0.852	0.805	0.825	0.655	0.709	0.763	0.763	0.605	0.720	0.801	0.809	0.813	0.785
	TI_4	0.886	0.895	**0.974**	**0.974**	0.880	0.882	0.959	0.961	0.883	0.882	0.931	0.934	0.833	0.842	0.810	0.803	0.733	0.789	0.829	0.842	0.892	0.882
	Avg	0.856	0.874	**0.939**	**0.942**	0.856	0.863	0.885	0.905	0.808	0.821	0.866	0.874	0.745	0.773	0.760	0.743	0.679	0.758	0.783	0.792	0.821	0.801
Action	TI_1	0.929	0.933	0.901	0.909	0.786	0.809	0.857	0.875	0.728	0.770	**0.921**	**0.923**	0.818	0.828	0.758	0.744	0.746	0.791	0.754	0.773	0.810	0.804
	TI_2	0.754	0.799	0.732	0.789	0.796	**0.818**	0.670	0.754	0.658	0.730	0.760	0.798	0.548	0.668	0.679	0.713	0.390	0.619	0.655	0.710	**0.800**	0.792
	TI_3	0.765	0.806	**0.813**	**0.841**	0.760	0.797	0.709	0.773	0.730	0.775	0.803	0.828	0.473	0.642	0.712	0.733	0.343	0.604	0.726	0.767	0.781	0.784
	TI_4	0.864	0.880	**0.849**	**0.865**	0.821	0.839	0.824	0.849	0.793	0.818	0.834	0.849	0.639	0.729	0.658	0.708	0.400	0.625	0.793	0.812	0.777	0.781
	Avg	0.828	0.854	0.824	**0.851**	0.791	0.816	0.765	0.813	0.727	0.773	**0.830**	0.849	0.620	0.717	0.702	0.724	0.470	0.660	0.732	0.766	0.792	0.790
Position	TI_1	0.921	0.922	**0.923**	**0.927**	0.791	0.814	0.895	0.904	0.705	0.757	0.901	0.903	0.823	0.837	0.748	0.732	0.719	0.778	0.736	0.756	0.780	0.758
	TI_2	0.672	0.751	0.732	0.789	**0.782**	**0.797**	0.695	0.768	0.653	0.715	0.690	0.750	0.516	0.659	0.649	0.681	0.359	0.607	0.637	0.694	0.762	0.742
	TI_3	0.706	0.770	0.766	**0.810**	0.757	0.791	0.687	0.760	0.718	0.760	0.775	0.807	0.455	0.630	0.709	0.725	0.347	0.606	0.697	0.746	**0.784**	0.780
	TI_4	0.753	0.800	**0.859**	**0.875**	0.747	0.790	0.826	0.850	0.751	0.795	0.832	0.850	0.627	0.715	0.743	0.765	0.305	0.590	0.655	0.705	0.782	0.780
	Avg	0.763	0.811	**0.820**	**0.850**	0.769	0.798	0.776	0.820	0.707	0.757	0.799	0.828	0.606	0.710	0.712	0.726	0.433	0.645	0.681	0.725	0.777	0.765
Scenario 2: Diverse Image Context																							
Object	TI_1	0.980	0.979	0.978	0.978	0.826	0.848	**0.980**	**0.980**	0.741	0.784	0.963	0.964	0.950	0.951	0.784	0.787	0.882	0.894	0.765	0.782	0.843	0.841
	TI_2	0.965	0.965	**0.960**	**0.961**	0.757	0.797	0.877	0.890	0.671	0.742	0.842	0.862	0.693	0.759	0.697	0.740	0.549	0.688	0.673	0.735	0.871	0.863
	TI_3	0.969	0.970	**0.963**	**0.964**	0.794	0.828	0.867	0.884	0.735	0.790	0.868	0.883	0.563	0.692	0.779	0.806	0.460	0.649	0.752	0.791	0.886	0.883
	TI_4	0.976	0.976	**0.962**	**0.962**	0.736	0.779	0.914	0.918	0.736	0.779	0.851	0.865	0.696	0.764	0.746	0.774	0.400	0.625	0.751	0.793	0.866	0.861
	Avg	0.972	0.972	**0.966**	**0.966**	0.778	0.813	0.909	0.918	0.721	0.774	0.881	0.893	0.726	0.792	0.751	0.777	0.573	0.714	0.735	0.775	0.867	0.862
Material	TI_1	0.952	0.954	**0.944**	**0.946**	0.846	0.854	0.934	0.936	0.758	0.795	0.926	0.929	0.885	0.886	0.768	0.758	0.800	0.829	0.792	0.804	0.782	0.762
	TI_2	0.889	0.898	**0.792**	**0.834**	0.752	0.772	0.758	0.804	0.655	0.723	0.743	0.784	0.579	0.688	0.681	0.693	0.395	0.621	0.712	0.745	0.749	0.722
	TI_3	0.915	0.920	**0.852**	**0.871**	0.713	0.759	0.796	0.832	0.682	0.741	0.776	0.809	0.465	0.643	0.675	0.686	0.302	0.588	0.792	0.807	0.741	0.722
	TI_4	0.925	0.930	**0.915**	**0.921**	0.771	0.794	0.856	0.869	0.631	0.710	0.750	0.785	0.570	0.682	0.693	0.706	0.260	0.575	0.756	0.771	0.652	0.640
	Avg	0.921	0.926	**0.876**	**0.893**	0.771	0.795	0.836	0.861	0.682	0.742	0.799	0.827	0.625	0.725	0.704	0.711	0.439	0.653	0.763	0.782	0.731	0.712
Color	TI_1	0.971	0.972	**0.960**	**0.961**	0.850	0.866	0.942	0.945	0.840	0.858	0.933	0.937	0.912	0.916	0.777	0.767	0.883	0.896	0.832	0.847	0.847	0.843
	TI_2	0.902	0.910	**0.869**	**0.883**	0.774	0.800	0.701	0.771	0.734	0.776	0.702	0.766	0.525	0.674	0.709	0.712	0.377	0.615	0.724	0.764	0.793	0.771
	TI_3	0.900	0.908	**0.904**	**0.911**	0.677	0.743	0.773	0.815	0.777	0.807	0.785	0.817	0.464	0.646	0.691	0.700	0.222	0.562	0.739	0.775	0.796	0.778
	TI_4	0.935	0.939	**0.946**	**0.947**	0.711	0.767	0.897	0.904	0.778	0.810	0.759	0.801	0.624	0.723	0.733	0.737	0.182	0.550	0.777	0.801	0.807	0.786
	Avg	0.927	0.932	**0.920**	**0.926**	0.753	0.794	0.828	0.859	0.782	0.813	0.795	0.830	0.631	0.740	0.727	0.729	0.416	0.656	0.768	0.797	0.811	0.795
Sentiment	TI_1	0.885	0.886	0.803	0.820	0.763	0.810	0.813	**0.845**	0.763	0.785	0.818	0.830	**0.825**	0.831	0.769	0.727	0.697	0.755	0.705	0.722	0.792	0.777
	TI_2	0.763	0.805	**0.848**	**0.860**	0.676	0.716	0.645	0.739	0.685	0.738	0.730	0.761	0.527	0.655	0.712	0.720	0.269	0.565	0.690	0.712	0.793	0.771
	TI_3	0.539	0.674	**0.862**	**0.867**	0.714	0.765	0.762	0.797	0.746	0.763	0.732	0.752	0.527	0.659	0.672	0.683	0.360	0.607	0.713	0.745	0.815	0.812
	TI_4	0.696	0.767	0.881	**0.883**	0.857	0.867	0.836	0.850	**0.885**	**0.883**	0.778	0.800	0.836	0.850	0.643	0.667	0.591	0.700	0.750	0.767	0.754	0.750
	Avg	0.721	0.783	**0.849**	**0.858**	0.752	0.789	0.764	0.808	0.770	0.792	0.764	0.786	0.679	0.749	0.699	0.699	0.479	0.657	0.715	0.736	0.789	0.778
Action	TI_1	0.944	0.946	**0.924**	**0.929**	0.818	0.836	0.879	0.891	0.761	0.797	0.914	0.917	0.824	0.835	0.753	0.749	0.771	0.811	0.780	0.795	0.831	0.826
	TI_2	0.658	0.743	0.697	**0.770**	0.704	0.760	0.474	0.660	0.563	0.675	0.667	0.740	0.337	0.591	0.597	0.662	0.168	0.545	0.609	0.702	**0.744**	0.756
	TI_3	0.759	0.806	0.756	0.809	0.645	0.728	0.617	0.727	0.631	0.719	**0.788**	**0.823**	0.283	0.577	0.651	0.696	0.120	0.532	0.631	0.709	0.762	0.776
	TI_4	0.814	0.843	**0.839**	**0.859**	0.626	0.722	0.753	0.798	0.688	0.748	0.778	0.813	0.458	0.641	0.639	0.692	0.114	0.530	0.726	0.768	0.787	0.798
	Avg	0.794	0.835	**0.804**	**0.842**	0.698	0.762	0.681	0.769	0.661	0.735	0.787	0.823	0.475	0.661	0.660	0.700	0.293	0.605	0.686	0.743	0.781	0.789
Position	TI_1	0.938	0.939	**0.927**	**0.931**	0.771	0.799	0.880	0.892	0.726	0.763	0.912	0.914	0.810	0.825	0.751	0.736	0.718	0.777	0.754	0.771	0.804	0.790
	TI_2	0.577	0.699	0.599	0.714	0.684	0.738	0.470	0.659	0.555	0.664	0.566	0.683	0.341	0.591	0.613	0.662	0.138	0.537	0.555	0.656	**0.748**	**0.744**
	TI_3	0.671	0.751	0.650	**0.741**	0.606	0.695	0.510	0.672	0.589	0.689	0.653	0.735	0.246	0.561	0.630	0.677	0.103	0.525	0.546	0.659	**0.728**	0.740
	TI_4	0.776	0.815	**0.832**	**0.855**	0.619	0.705	0.721	0.775	0.663	0.720	0.722	0.765	0.400	0.610	0.619	0.655	0.095	0.525	0.715	0.765	0.731	0.720
	Avg	0.740	0.801	0.752	**0.810**	0.670	0.734	0.645	0.750	0.633	0.709	0.713	0.774	0.449	0.647	0.653	0.682	0.264	0.591	0.643	0.713	**0.753**	0.748

Note: **Bold** and underlined denote best and second-best results (excluding GPT-4o).

occurrences across diverse images without hallucinating. Alternatively, models might possess an inherent bias towards outputting No when uncertain, which artificially boosts accuracy on "each" queries (where No is the strictly correct answer if the targeted item is absent in even a single image).



(a) InternVL-8B (UIC) (b) InternVL-8B (DIC) (c) LLaVA-IL (UIC) (d) LLaVA-IL (DIC)

(e) LLaVA-OV (UIC) (f) LLaVA-OV (DIC) (g) Mantis-8B (UIC) (h) Mantis-8B (DIC)

(i) MiniCPM-4.5 (UIC) (j) MiniCPM-4.5 (DIC) (k) Qwen3-32B (UIC) (l) Qwen3-32B (DIC)

(m) Qwen2.5-7B (UIC) (n) Qwen2.5-7B (DIC) (o) Qwen3-8B (UIC) (p) Qwen3-8B (DIC)

(q) GPT-4o (UIC) (r) GPT-4o (DIC)

Figure 2: Full multi-granularity evaluation results on LIVE for the remaining nine LVLMs at a temperature of 0. For each model, we report accuracy under Uniform Image Context (UIC, left) and Diverse Image Context (DIC, right).

Table 5: Intra-group CLIP Similarity.

Setting	Obj	Mat	Col	Sen	Act	Pos	Avg
UIC	0.65	0.64	0.65	0.62	0.65	0.65	**0.65**
DIC	0.54	0.55	0.55	0.52	0.56	0.56	**0.55**

❷ **Varying Model Sensitivities.** The magnitude of this logical gap varies considerably between architectures. For instance, at deterministic decoding ($t = 0$), Qwen2.5-7B and MiniCPM-2.6 exhibit massive gaps of -0.335 and -0.288 respectively, indicating a severe vulnerability to existential formulations. Conversely, the proprietary GPT-4o shows a remarkably narrow gap (-0.029 at $t = 0$ and -0.026 at $t = 0.7$), demonstrating highly robust and balanced logical reasoning capabilities regardless of the quantifier used.

Table 6: Performance comparison between "Any" and "Each" logical formulations across different temperatures. Gap is calculated as ($\mathbf{Acc_{Any}} - \mathbf{Acc_{Each}}$).

Model	Temperature = 0			Temperature = 0.7			Temperature = 1.0		
	Any	Each	Gap	Any	Each	Gap	Any	Each	Gap
GPT-4o	0.873	0.983	-0.110	0.874	0.987	-0.113	0.872	0.982	-0.110
Qwen3-32B	0.871	0.981	-0.110	0.871	0.981	-0.110	0.872	0.980	-0.108
InternVL-26B	0.833	0.933	-0.100	0.809	0.905	-0.096	0.790	0.892	-0.102
Qwen3-8B	0.808	0.957	-0.149	0.811	0.957	-0.146	0.813	0.957	-0.144
InternVL-8B	0.808	0.912	-0.104	0.790	0.853	-0.063	0.767	0.846	-0.079
MiniCPM-4.5	0.827	0.968	-0.142	0.820	0.957	-0.137	0.809	0.952	-0.143
MiniCPM-2.6	0.662	0.947	-0.285	0.680	0.927	-0.247	0.680	0.911	-0.231
Mantis-8B	0.819	0.871	-0.052	0.758	0.823	-0.065	0.720	0.796	-0.076
Qwen2.5-7B	0.588	0.921	-0.332	0.602	0.902	-0.300	0.612	0.882	-0.270
LLaVA-OV	0.845	0.906	-0.061	0.789	0.811	-0.022	0.756	0.791	-0.035
LLaVA-IL	0.882	0.975	-0.093	0.825	0.952	-0.127	0.785	0.919	-0.134

❸ **Temperature Robustness.** While increasing the temperature introduces more variance, the directional trend ($\text{Acc}_{\text{Each}} > \text{Acc}_{\text{Any}}$) remains strictly consistent across all models and settings, validating that this is an intrinsic cognitive limitation of current LVLM architectures rather than a decoding artifact.

9 Extended Evaluation on Larger Image Subsets ($k = 5$ to 10)

To further investigate Large Vision-Language Models' (LVLMs) multi-image understanding capabilities under extended contextual loads, we supplement our primary evaluation with an assessment on larger image subsets ranging from $k = 5$ to $k = 10$ images.

We adopt the identical rigorous anchor-based generation and filtering pipeline utilized for the 4-image configurations in the main paper. However, to maintain computational tractability and manage API costs associated with the combinatorial explosion of larger sets, we adapt our multi-granularity sampling strategy. Instead of exhaustive combinatorial sampling, we randomly select subsets of targeted images for questioning. To ensure statistical significance and evaluation rigor, we maintain exactly 3,000 balanced yes/no question-answer pairs for each k-image configuration. These questions are strictly balanced across the six visual recognition tasks. For conciseness and ease of comparison, we report the averaged F1 scores and accuracy across all six tasks.

The aggregated results, presented in Table 7, yield several key observations that systematically align with and extend our core findings:

❶ **Continuous Performance Degradation with Increased Context.** Models generally suffer from performance degradation as the number of images increases from 5 to 10. For instance, in UIC scenarios, Qwen3-32B's F1 score smoothly degrades from 0.914 at $k = 5$ down to 0.894 at $k = 10$. Similar vulnerability to extended visual context is evident in smaller models like Qwen2.5-7B, which sees its UIC F1 score drop from 0.736 to 0.710. This indicates that tracking factual grounding and managing cognitive load across an expanding sequence of visual contexts remains a challenging open problem for current architectures.

❷ **Persistent Context Interference in DIC Scenarios.** Reaffirming our findings from the main text, Diverse Image Contexts (DIC) consistently induce higher hallucination rates than Uniform Image Contexts (UIC) across larger image sets. For example, at $k = 6$, Qwen3-8B achieves an F1 of 0.910 in UIC but drops notably to 0.865 in DIC. Similarly, at the extreme $k = 10$ setting, Qwen2.5-7B exhibits a severe drop from an F1 of 0.710 in UIC to 0.573 in DIC. This confirms that semantic heterogeneity persistently exacerbates context interference and cross-image attention confusion, regardless of the overall sequence length.

10 Extension Beyond Binary Questioning

While balanced binary (Yes/No) questions establish a rigorous baseline for assessing hallucination rates, modern advanced LVLMs necessitate more challenging evaluation formats to fully expose their cognitive boundaries. Leveraging the structured visual information extracted in Stage 2 and the robust instruction-following capabilities of LLMs (specifically Qwen2.5-7B-Instruct), we seamlessly extend our LIVE pipeline to support three new advanced question paradigms. To ensure robust and efficient evaluation, we constructed 3,000 samples for each format, strictly maintaining a balanced distribution across the six visual recognition tasks. The evaluation results across these new formats are presented in Figure 5.

❶ **Image Identification ("Which Image(s)"):** To explicitly test multi-granularity localization and mitigate the 50% random-guess baseline of binary questions, we transform existence queries into retrieval tasks. For a targeted multi-image real item or counterfactual item, the prompt is formulated as: *"Among the given 4 images, which image(s) contain [⋆]? Please list their indices (e.g., 1, 2) or state 'None'."* This format directly probes content confusion in UIC scenarios, as models hallucinating features across similar images will erroneously append incorrect image indices. Evaluation is efficiently conducted via regular expression matching against the ground-truth sets.

❷ **Multiple-Choice QA:** To combine the rigorousness of open-ended queries with the standardized evaluation efficiency of binary tasks, we introduce a multiple-choice format. The LVLM is provided with an exclusion question (e.g., *"Which of the following is NOT present in any of image X and image Y?"*) alongside four options (A, B, C, D). Crucially, the distractors are deliberately sampled from the counterfactual hallucination items generated during Stage 2. This format provides a highly discriminative assessment of multi-image hallucination while allowing for strict, unambiguous evaluation via exact option-letter matching.

❸ **Open-Ended QA:** To evaluate nuanced visual comprehension beyond simple presence/absence, we design open-ended factual questions that require models to describe specific visual properties (e.g., *"What type of road is visible in any of image 1 and image 4?"*). Because free-form text generation introduces significant variations in phrasing, we employ an efficient LLM-as-a-judge protocol (utilizing Qwen2.5-7B-Instruct) to conduct fast, concurrent, and accurate assessments of the predicted reasoning against the concise ground-truth answers.

11 More Case Studies

To further illustrate the diverse evaluation scenarios in the LIVE benchmark, this section provides comprehensive case studies across

Table 7: Average performance across different k values ($k \in [5, 10]$). Metrics are averaged across all six visual recognition tasks to provide an overall assessment of multi-image understanding under extended contexts.

Task	GPT-4o		Qwen3-32B		InternVL-26B		Qwen3-8B		InternVL-8B		MiniCPM-4.5		MiniCPM-2.6		Mantis-8B		Qwen2.5-7B		LLaVA-OV		LLaVA-IL	
MAP	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc
Scenario 1: Uniform Image Context (UIC)																						
k5	0.872	0.873	**0.914**	**0.910**	0.865	0.864	0.902	0.902	0.856	0.869	0.896	0.892	0.869	0.876	0.814	0.808	0.736	0.800	0.884	0.882	0.849	0.820
k6	0.867	0.871	**0.915**	**0.914**	0.899	0.897	0.910	0.908	0.860	0.872	0.904	0.904	0.876	0.885	0.818	0.814	0.763	0.813	0.874	0.874	0.834	0.806
k7	0.870	0.875	**0.899**	**0.901**	0.841	0.843	0.893	0.896	0.835	0.848	0.891	0.891	0.841	0.854	0.760	0.757	0.721	0.791	0.836	0.841	0.830	0.799
k8	0.877	0.881	**0.913**	**0.913**	0.849	0.849	0.896	0.896	0.834	0.855	0.891	0.891	0.844	0.860	0.752	0.756	0.713	0.792	0.863	0.865	0.829	0.792
k9	0.867	0.871	0.873	**0.876**	0.846	0.847	0.861	0.869	0.833	0.852	**0.874**	0.872	0.832	0.849	0.771	0.767	0.677	0.770	0.829	0.838	0.816	0.778
k10	0.876	0.875	0.894	0.889	0.822	0.836	0.883	0.887	0.811	0.841	**0.897**	**0.899**	0.821	0.846	0.745	0.745	0.710	0.785	0.800	0.817	0.801	0.757
Scenario 2: Diverse Image Context (DIC)																						
k5	0.886	0.889	0.897	0.891	0.875	0.878	0.889	0.885	0.834	0.854	**0.900**	**0.899**	0.848	0.863	0.756	0.767	0.664	0.765	0.840	0.848	0.816	0.795
k6	0.858	0.866	0.864	0.887	0.858	0.858	0.865	0.888	0.787	0.830	**0.890**	**0.903**	0.791	0.831	0.716	0.752	0.592	0.737	0.811	0.838	0.826	0.800
k7	0.826	0.855	**0.900**	0.897	0.826	0.839	0.863	0.874	0.842	0.857	0.896	**0.898**	0.861	0.875	0.737	0.761	0.694	0.777	0.847	0.851	0.821	0.791
k8	0.863	0.873	**0.907**	**0.903**	0.847	0.852	0.883	0.877	0.820	0.840	0.902	0.901	0.852	0.865	0.692	0.715	0.636	0.743	0.830	0.829	0.794	0.752
k9	0.852	0.864	0.866	0.879	0.828	0.842	0.867	0.873	0.775	0.818	**0.875**	**0.884**	0.771	0.812	0.701	0.722	0.590	0.726	0.777	0.804	0.792	0.769
k10	0.832	0.856	**0.886**	**0.884**	0.832	0.842	0.864	0.857	0.802	0.835	0.867	0.865	0.747	0.803	0.660	0.693	0.573	0.721	0.792	0.814	0.816	0.794

Note: **Bold** and underlined denote best and second-best results (excluding GPT-4o), respectively.

various cognitive tasks and context scales. As introduced in Section 10, modern LVLMs are evaluated not only on binary reasoning but also on extended formats to rigorously expose their cognitive boundaries.

In the following figures, we showcase the model performances under both Uniform Image Context (UIC) and Diverse Image Context (DIC). **Figure 6**, **Figure 7**, and **Figure 8** present the qualitative results for the Image Identification, Multiple-Choice QA, and Open-Ended QA tasks, respectively. These formats directly probe the models' ability to mitigate context interference and content confusion without relying on random guessing. Furthermore, from **Figure 9** to **Figure 16**, we present a large-scale evaluation of object hallucination spanning from standard $k = 4$ settings up to challenging $k = 10$ image contexts. To ensure optimal visual alignment, the cases are systematically grouped into 2×2 grids by their image aspect ratios. These examples visually demonstrate how hallucination rates and cross-image interference amplify as the visual cognitive load increases.

(a) InternVL-26B (UIC) (b) InternVL-26B (DIC) (c) InternVL-8B (UIC) (d) InternVL-8B (DIC)

(e) LLaVA-IL (UIC) (f) LLaVA-IL (DIC) (g) LLaVA-OV (UIC) (h) LLaVA-OV (DIC)

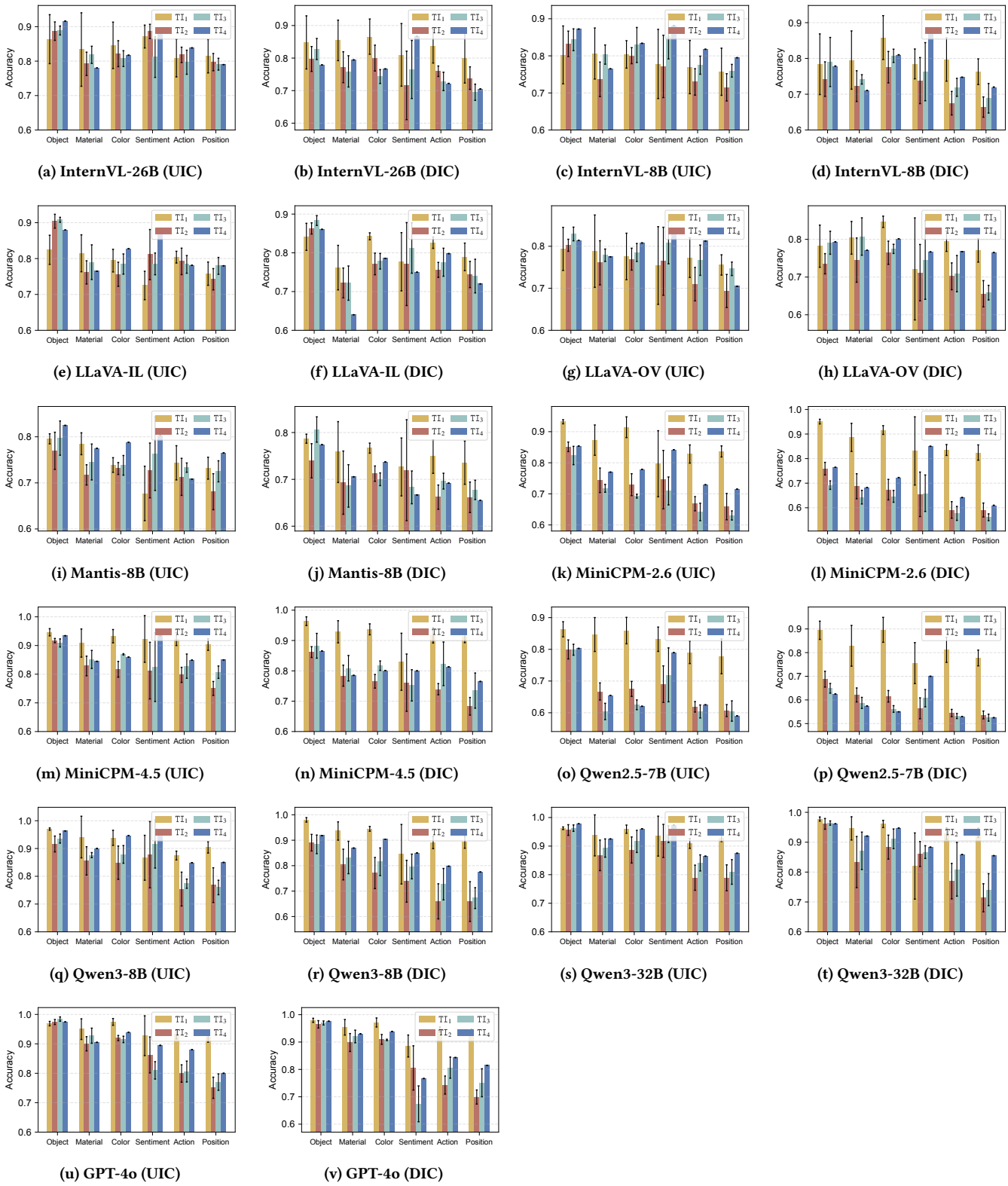
(i) Mantis-8B (UIC) (j) Mantis-8B (DIC) (k) MiniCPM-2.6 (UIC) (l) MiniCPM-2.6 (DIC)

(m) MiniCPM-4.5 (UIC) (n) MiniCPM-4.5 (DIC) (o) Qwen2.5-7B (UIC) (p) Qwen2.5-7B (DIC)

(q) Qwen3-8B (UIC) (r) Qwen3-8B (DIC) (s) Qwen3-32B (UIC) (t) Qwen3-32B (DIC)

(u) GPT-4o (UIC) (v) GPT-4o (DIC)

Figure 3: Full multi-granularity evaluation results on LIVE for all 11 LVLMs at a temperature of 0.7. For each model, we report accuracy under Uniform Image Context (UIC, left) and Diverse Image Context (DIC, right).



(a) InternVL-26B (UIC) (b) InternVL-26B (DIC) (c) InternVL-8B (UIC) (d) InternVL-8B (DIC)

(e) LLaVA-IL (UIC) (f) LLaVA-IL (DIC) (g) LLaVA-OV (UIC) (h) LLaVA-OV (DIC)

(i) Mantis-8B (UIC) (j) Mantis-8B (DIC) (k) MiniCPM-2.6 (UIC) (l) MiniCPM-2.6 (DIC)

(m) MiniCPM-4.5 (UIC) (n) MiniCPM-4.5 (DIC) (o) Qwen2.5-7B (UIC) (p) Qwen2.5-7B (DIC)

(q) Qwen3-8B (UIC) (r) Qwen3-8B (DIC) (s) Qwen3-32B (UIC) (t) Qwen3-32B (DIC)

(u) GPT-4o (UIC) (v) GPT-4o (DIC)

Figure 4: Full multi-granularity evaluation results on LIVE for all 11 LVLMs at a temperature of 1.0. For each model, we report accuracy under Uniform Image Context (UIC, left) and Diverse Image Context (DIC, right).

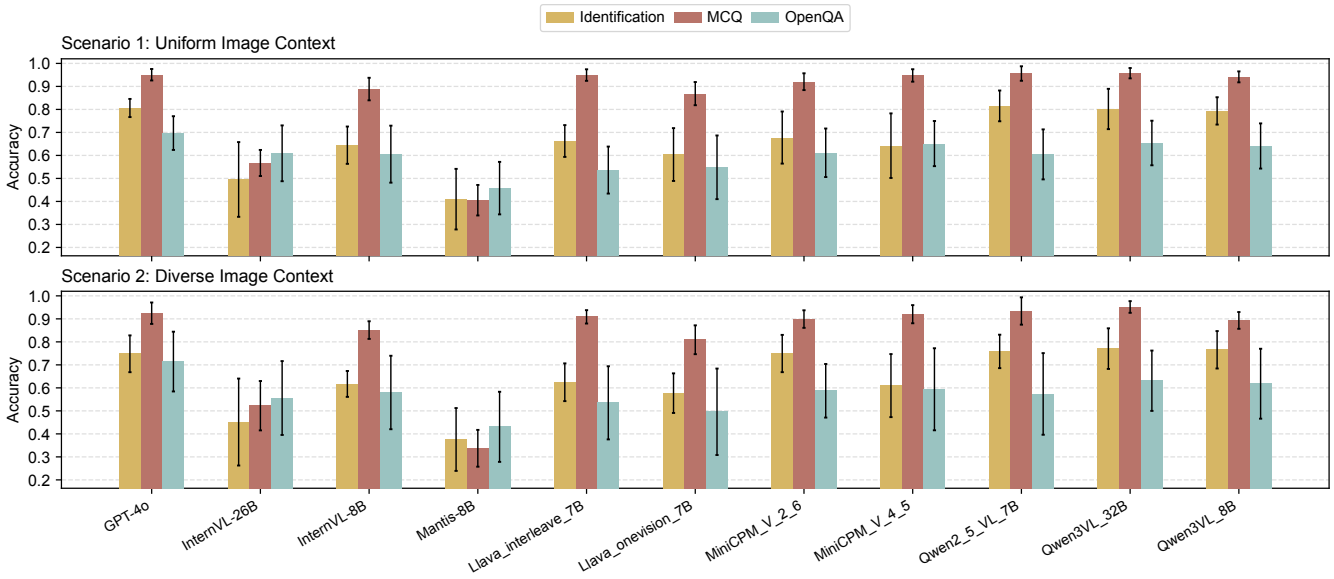


Figure 5: Performance evaluation of LVLMs across three advanced formatting paradigms: Image Identification, Multiple-Choice QA, and Open-Ended QA. Metrics are averaged across six visual recognition tasks.

Task: Action **Scenario:** UIC **Model:** GPT4o

Question: Among the given 4 images, which image(s) contain person eating at table? Please list their indices (e.g., 1, 2) or state 'None'.

Ground Truth: 1, 3

Model Response: 1, 2, 3
Reasoning: In image 1, a hand holding a fork is visible, indicating someone is eating at the table. In image 2, a hand is visible near the sandwich, suggesting someone is eating at the table. In image 3, a person is visible in the background eating at the table.

Task: Material **Scenario:** UIC **Model:** Qwen2-5-VL-7B

Question: Among the given 4 images, which image(s) contain a table made of wood? Please list their indices (e.g., 1, 2) or state 'None'.

Ground Truth: 4

Model Response: 1,4
Reasoning: In Image 1, there is a parking meter with a sign indicating it accepts only quarters and dollar coins, but no table is visible. In Image 4, there is a wooden table where a doll is sitting and drawing.

Task: Object **Scenario:** UIC **Model:** Qwen3VL-8B

Question: Among the given 4 images, which image(s) contain a table made of wood? Please list their indices (e.g., 1, 2) or state 'None'.

Ground Truth: 1

Model Response: 1,4
Reasoning: Image 1 depicts a woman seated at a table with a pizza in what appears to be a restaurant, evidenced by the menu stands, wall decor, and ambient lighting typical of a dining establishment. Image 4 shows three women cutting a cake, likely at a celebration, and the setting includes a table with a cake, candles, and a decorative candlestick, which is consistent with a restaurant or banquet hall environment. Image 2 is clearly a kitchen or home setting with an oven and wheelchair. Image 3 is a party scene, but the background and setup (e.g., tiered stand with pastries, specific decor) suggest a private home or event space rather than a restaurant Therefore, only images 1 and 4 contain restaurant-like settings.

Task: Object **Scenario:** DIC **Model:** Qwen3-VL-8B

Question: Among the given 4 images, which image(s) contain bird? Please list their indices (e.g., 1, 2) or state 'None'.

Ground Truth: 2

Model Response: None
Reasoning: I have examined all four images to determine if any contain a bird.
Image 1: Shows a man in a bathroom with a small dog. No birds are visible.
Image 2: Shows two dogs on a pebble beach. No birds are visible.
Image 3: Shows a toilet in what appears to be a ship's head (bathroom). No birds are visible.
Image 4: Shows a long-haired cat sitting inside an open suitcase. No birds are visible.
None of the images contain a bird. Therefore, the correct answer is "None".

Figure 6: Case study under UIC and DIC scenarios (Image Identification). denotes factual errors, denotes conflicts with target image, and denote correct target image details.

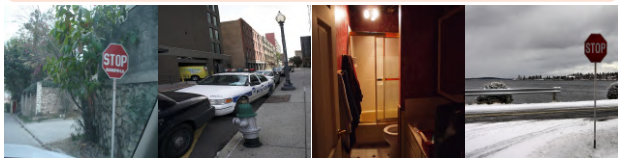
Task: Color **Scenario:** UIC **Model:** LlaVA-onevision-7B

Question: Which of the following is NOT present in any of image 2 and image 3? (Options: A: fridge is blue B: white mini fridge C: colorful labels on fridge D: dirty dishes on table)

Ground Truth: A

Model Response: B
Reasoning: The image does not show a white mini fridge in either image 2 or image 3.

Task: Position **Scenario:** DIC **Model:** LLaVA-onevision-7B

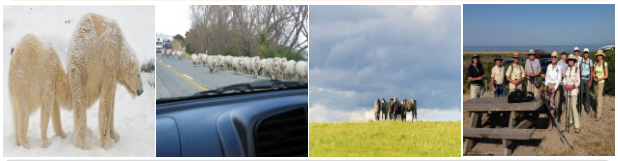


Question: Which of the following is NOT present in any of image 2 and image 4? (Options: A: stop sign lying flat on ground B: police car parked on side of road C: fire hydrant near curb D: stop sign standing next to snowy road)

Ground Truth: A

Model Response: no
Reasoning: The stop sign in image 4 is standing upright and not lying flat on the ground, unlike the one in image 2.

Task: Color **Scenario:** DIC **Model:** Qwen2.5-VL-7B



Question: What color is the interior in any of image 2 and image 4? (Options: A: interior is blue B: interior is gray C: interior is beige D: interior is black)

Ground Truth: A

Model Response: B
Reasoning: In image 4, the interior of the car visible through the windshield appears to be gray, as indicated by the color of the dashboard and the air vent.

Task: Action **Scenario:** UIC **Model:** GPT-4o

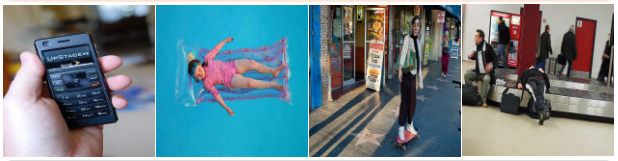
Question: Which of the following is NOT happening in any of image 2 and image 3? (Options: A: People are avoiding all traffic and walking on the sidewalks. B: A bus is driving down the road. C: Pedestrians are crossing a bridge. D: Cars are moving through busy traffic.)

Ground Truth: A

Model Response: C
Reasoning: In image 2, a bus is driving down the road, and cars are moving through busy traffic, so options B and D are happening. In image 3, a bus is crossing a bridge, but there are no visible pedestrians crossing the bridge, so option C is not happening. Additionally, in image 2, people are walking on the sidewalks, so option A is also happening. Therefore, the correct answer is C.

Figure 7: Case study under UIC and DIC scenarios (Multiple-Choice QA). denotes factual errors, denotes conflicts with target image, and denote correct target image details.

Task: Object **Scenario:** DIC **Model:** Qwen3-VL-32B



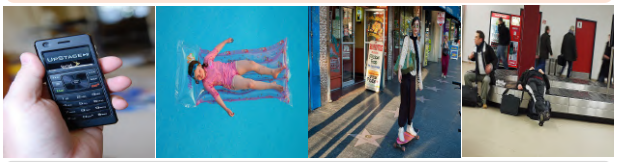
Question: What material is the platform made from in any of image 1, image 2 and image 4?

Ground Truth: metal

Model Response: Concrete

Reasoning: In image 3, the platform (sidewalk) is made of concrete, as evidenced by its gray, textured surface and the embedded star-shaped plaques typical of the Hollywood Walk of Fame. Images 1 and 2 do not show a platform; image 1 shows a hand holding a phone, and image 2 shows a child floating in a pool. Image 4 shows a baggage carousel, not a platform. Therefore, only image 3 contains a platform, and it is made of concrete.

Task: Material **Scenario:** DIC **Model:** Qwen3-VL-32B



Question: What material is the platform made from in any of image 1, image 2 and image 4?

Ground Truth: metal

Model Response: Concrete

Reasoning: In image 3, the platform (sidewalk) is made of concrete, as evidenced by its gray, textured surface and the embedded star-shaped plaques typical of the Hollywood Walk of Fame. Images 1 and 2 do not show a platform; image 1 shows a hand holding a phone, and image 2 shows a child floating in a pool. Image 4 shows a baggage carousel, not a platform. Therefore, only image 3 contains a platform, and it is made of concrete.

Task: Action **Scenario:** UIC **Model:** InternVL2.5-8B

Question: What color top is the girl wearing on her upper body in any of image 2 and image 3?

Ground Truth: white and red

Model Response: White both

Reasoning: White both image 2 and image 3, the girl is wearing a white top.

Task: Material **Scenario:** UIC **Model:** Qwen3-VL-8B

Question: What material is the food tray made of in image 3?

Ground Truth: plastic

Model Response: Aluminum foil

Reasoning: In image 3, the food tray contains multiple compartments lined with aluminum foil, which is clearly visible around the rice, curry, and broth. The foil is also used for the wrapped item and the condiment packet, indicating it's the material of the tray.

Figure 8: Case study under UIC and DIC scenarios (Open-Ended QA). denotes factual errors, denotes conflicts with target image, and denote correct target image details.

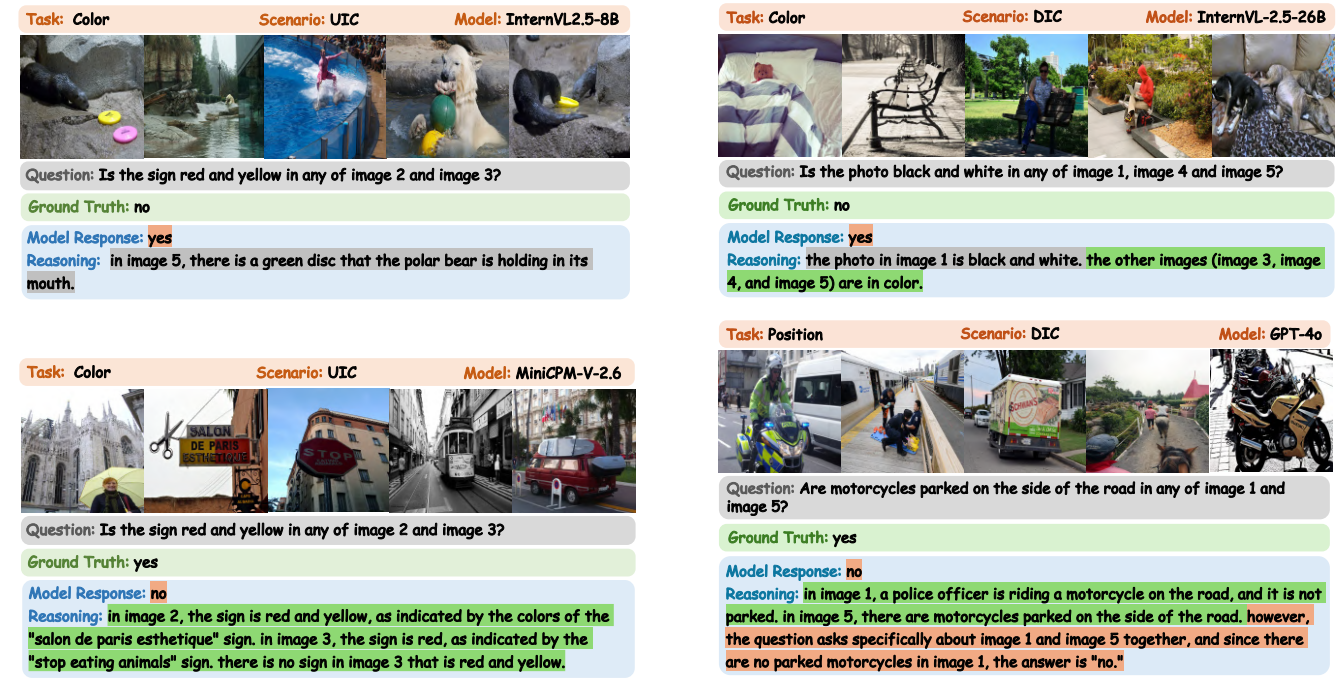
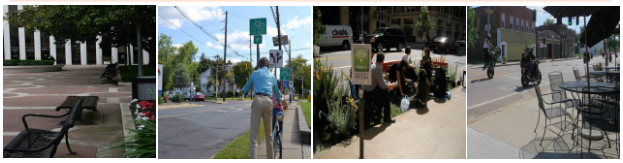


Figure 9: Case study under UIC and DIC scenarios (Scaling contexts $k \geq 4$). Grouped by aspect ratio to ensure visual alignment. denotes factual errors, denotes conflicts with target image, and denote correct target image details.

Task: Object **Scenario:** UIC **Model:** InternVL-26B



Question: Is there a child in any of image 2 and image 4?

Ground Truth: yes

Model Response: no
Reasoning: there is no child visible in either image 2 or image 4. image 2 shows adults sitting on a bench, and image 4 shows adults on motorcycles.

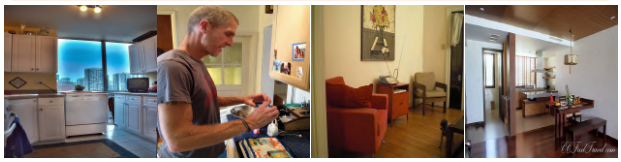
Task: Position **Scenario:** UIC **Model:** Qwen3VL-32B

Question: Is there a car in a parking lot in any of image 1, image 3, image 4, image 2 and image 5?

Ground Truth: no

Model Response: yes
Reasoning: in image 1, there is a teal-colored car parked on the side of a residential street, which can be considered a parking area. the car is positioned next to a curb, with other vehicles visible in the background, indicating a typical street parking scenario. while images 2, 3, 4, and 5 show outdoor settings (a park, a rural path, a road with spectators, and a skatepark respectively), none of them clearly depict a formal parking lot. however, image 1 does show a car in a parking context, satisfying the condition. therefore, the answer is "yes."

Task: Object **Scenario:** DIC **Model:** GPT-4o



Question: Is there a sitting area in any of image 1 and image 3?

Ground Truth: yes

Model Response: no
Reasoning: image 1 shows a kitchen with cabinets, appliances, and a window view of city buildings. there's no sitting area visible. image 3 shows what appears to be a modern open-plan space with a dining area (table with benches) and kitchen visible, but no dedicated sitting area with sofas or lounge seating.

Task: Material **Scenario:** DIC **Model:** GPT-4o

Question: Is the road made of bricks in any of image 2, image 4 and image 6?

Ground Truth: yes

Model Response: no
Reasoning: in image 2, the road is made of asphalt or a similar material, not bricks. in image 4, the ground is made of bricks, as seen around the painted fire hydrant. in image 6, the path appears to be paved with a smooth material, not bricks.

Figure 10: Case study under UIC and DIC scenarios (Scaling contexts $k \geq 4$). denotes factual errors, denotes conflicts with target image, and denote correct target image details.

Task: Material **Scenario:** DIC **Model:** GPT-4o

Question: Is the road made of bricks in any of image 2, image 4 and image 6?

Ground Truth: yes

Model Response: no

Reasoning: in image 2, the road is made of asphalt or a similar material, not bricks. in image 4, the ground is made of bricks, as seen around the painted fire hydrant. in image 6, the path appears to be paved with a smooth material, not bricks.

Task: Action **Scenario:** UIC **Model:** MiniCPM-V-4.5

Question: Is a person sitting on the ground in any of image 3, image 5, image 6, image 8 and image 9?

Ground Truth: no

Model Response: yes

Reasoning: in image 3, a person is sitting on the ground near the water, surrounded by birds.

Task: Object **Scenario:** DIC **Model:** MiniCPM-V-4.5

Question: Is there a baseball pitcher in any of image 2, image 3 and image 4?

Ground Truth: yes

Model Response: no

Reasoning: in image 2, the individual is a baseball player holding a bat, not a pitcher. in image 3, the individuals are shown in a sequence of pitching actions, but they are not a single pitcher. in image 4, the individuals are dressed in formal attire, not in baseball uniforms, and there is no baseball pitcher present.

Task: Position **Scenario:** DIC **Model:** Qwen3-VL-32B

Question: Is the man standing in front of the mirror in any of image 4 and image 5?

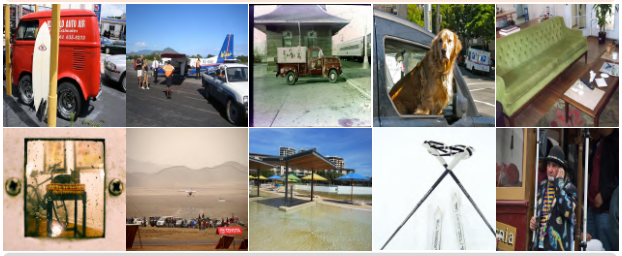
Ground Truth: yes

Model Response: no

Reasoning: in image 4, a man is standing in front of a mirror in a bathroom, brushing his teeth and holding a red object (likely a phone or hairbrush) to his head. the mirror reflects his image, confirming his presence in front of it. in image 5, the object shown is a close-up of a round, metallic, reflective surface—likely a magnifying mirror or part of a device—but there is no person visible in the image, and no one is standing in front of it. therefore, while image 4 clearly shows a man in front of a mirror, image 5 does not. the question asks if the man is standing in front of the mirror in *any* of image 4 and image 5. since image 5 does not show a man at all, the answer is "no" because the condition is not met in both images.

Figure 11: Case study under UIC and DIC scenarios (Scaling contexts $k \geq 4$). denotes factual errors, denotes conflicts with target image, and denote correct target image details.

Task: Action **Scenario:** DIC **Model:** MiniCPM-V-2.6



Question: Is a dog looking out of a car window in any of image 4 and image 7?

Ground Truth: yes

Model Response: no
Reasoning: in image 4, a dog is indeed looking out of a car window, as evidenced by the dog's position and gaze directed outside the vehicle. however, in image 7, there is no car window visible, and the focus is on a cat sitting on a stool, not a dog looking out of a window.

Task: Color **Scenario:** UIC **Model:** MiniCPM-V-2.6

Question: Is the light green in any of image 2, image 4, image 5, image 8 and image 9?

Ground Truth: yes

Model Response: no
Reasoning: in image 2, the traffic light is not green; it is red. in image 4, the traffic light is not green; it is red as well. in image 5, there is no traffic light visible. in image 8, the traffic light is not green; it is red. in image 9, the traffic light is not green; it is red. therefore, the light green is not present in any of the specified images.

Task: Color **Scenario:** DIC **Model:** InternVL-2.5-26B

Question: Is the shirt black in any of image 4 and image 8?

Ground Truth: yes

Model Response: no
Reasoning: in image 4, the man is wearing a black shirt, and in image 8, the man is wearing a white shirt with a black tie.

Task: Object **Scenario:** UIC **Model:** MiniCPM-V-4.5

Question: Is the ocean present in any of image 6 and image 7?

Ground Truth: no

Model Response: yes
Reasoning: in image 6, the ocean is visible in the background, and in image 7, the ocean is also visible in the background.

Figure 12: Case study under UIC and DIC scenarios (Scaling contexts $k \geq 4$). denotes factual errors, denotes conflicts with target image, and denote correct target image details.

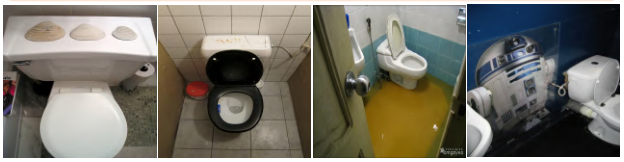
Task: Color **Scenario:** DIC **Model:** Qwen3-VL-8B

Question: Is the clock black in any of image 2 and image 3?

Ground Truth: no

Model Response: yes
Reasoning: in image 2, the clock face is black. the numbers and hands on the clock face are black, set against a gold or brass-colored background. the outer casing of the clock is dark wood, but the face itself is black.
in image 3, the clock face is also black. the numbers and hands are black, set against a white background. the clock is a digital clock, but the display is black with white numbers and symbols. the clock face is black.
therefore, the clock is black in both image 2 and image 3.

Task: Object **Scenario:** UIC **Model:** Qwen3-VL-32B



Question: Is there a wall paper in any of image 1, image 2 and image 4?

Ground Truth: yes

Model Response: no
Reasoning: - **image 1**: the walls are covered in white tiles, not wallpaper. - **image 2**: the walls are also covered in white tiles, with no visible wallpaper. - **image 4**: the wall features a large, printed image of r2-d2, which appears to be a decorative panel or vinyl decal, not traditional wallpaper. while it is a decorative surface covering, it is not classified as wallpaper in the conventional sense (which typically refers to paper-based, patterned wall coverings). therefore, none of the images show traditional wallpaper.

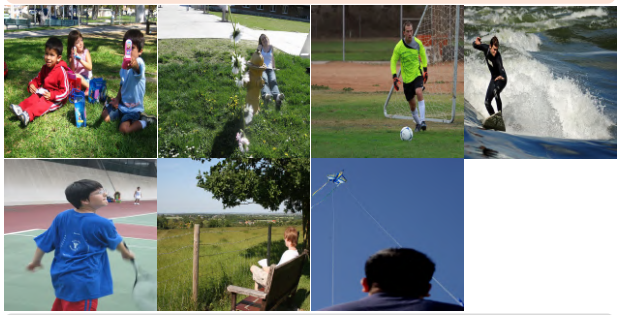
Task: Position **Scenario:** DIC **Model:** GPT-4o

Question: Is a skateboard being ridden in any of image1 and image7?

Ground Truth: yes

Model Response: no
Reasoning: in image 1, there is no skateboard being ridden; the individuals are sitting on a statue in a garden. in image 7, a skateboard is being ridden by a person on a smooth surface.

Task: Action **Scenario:** DIC **Model:** InternVL-2.5-8B



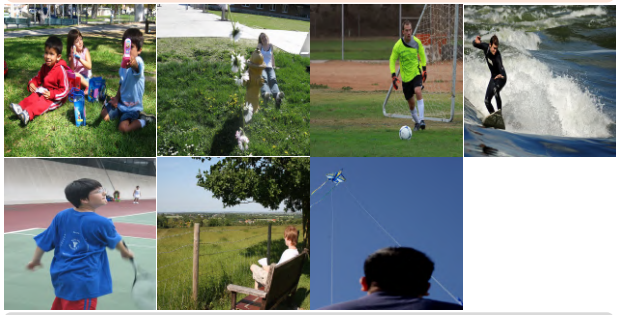
Question: Is the girl wearing a white tank top in any of image 1 and image 2?

Ground Truth: yes

Model Response: no
Reasoning: in image 1, the girl is wearing a pink top, and in image 2, the girl is wearing a white tank top. these are distinct clothing items, so the answer is no.

Figure 13: Case study under UIC and DIC scenarios (Scaling contexts $k \geq 4$). denotes factual errors, denotes conflicts with target image, and denote correct target image details.

Task: Action **Scenario:** DIC **Model:** InternVL-2.5-8B



Question: Is the girl wearing a white tank top in any of image 1 and image 2?

Ground Truth: yes

Model Response: no
Reasoning: in image 1, the girl is wearing a pink top, and in image 2, the girl is wearing a white tank top. these are distinct clothing items, so the answer is no.

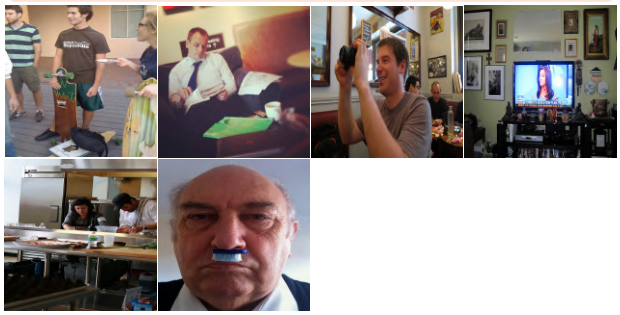
Task: Position **Scenario:** UIC **Model:** GPT4o

Question: Is the umpire kneeling on the field in any of image 3, image 5 and image 6?

Ground Truth: no

Model Response: yes
Reasoning: in **image 6**, the umpire is kneeling on the field behind the catcher. in **image 3** and **image 5**, there is no umpire visible kneeling on the field.

Task: Sentiment **Scenario:** DIC **Model:** GPT-4o

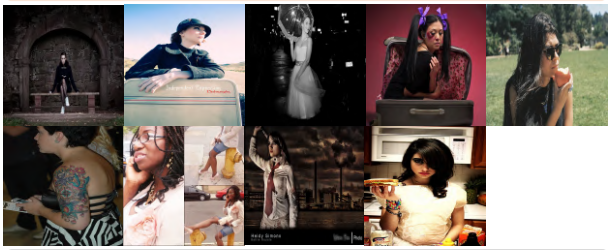


Question: Is the man smiling in any of image 3 and image 5?

Ground Truth: yes

Model Response: no
Reasoning: in image 3, the man appears to be holding a camera and does not show any visible signs of smiling. in image 5, the individuals are focused on their tasks in the kitchen, and there are no clear indications of a smile.

Task: Sentiment **Scenario:** UIC **Model:** MiniCPM-V-2.6



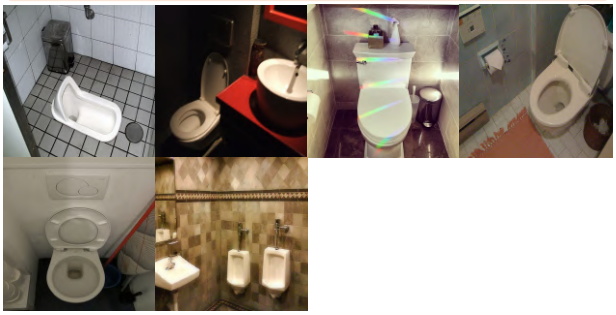
Question: Does the lady appear happy or content in any of image 1, image 2, image 3, image 5, image 6, image 7, image 8 and image 9?

Ground Truth: yes

Model Response: no
Reasoning: in all the images provided, the lady does not display clear signs of happiness or contentment. in image 1, she has a neutral expression. in image 2, she is looking away with a contemplative expression. in image 3, she is smiling, but the context of the image suggests a more dramatic or staged scenario. in image 5, she is focused on eating, and in image 6, she is looking away from the camera, which does not convey happiness. in image 7, she is engaged in a phone conversation, and in image 8, she is in a dramatic pose with a prop, which does not indicate contentment. in image 9, she is holding a sandwich, which is a neutral action, and her expression is not one of happiness.

Figure 14: Case study under UIC and DIC scenarios (Scaling contexts $k \geq 4$). denotes factual errors, denotes conflicts with target image, and denote correct target image details.

Task: Material **Scenario:** UIC **Model:** InternVL2.5-26B



Question: Is the toilet made of a non-ceramic material in any of image 1, image 2, image 3, image 4, image 5 and image 6?

Ground Truth: no

Model Response: yes
Reasoning: in image 1, the toilet is made of a non-ceramic material, as it appears to be a squat toilet, which is typically made of plastic or other non-ceramic materials. the other images show toilets that are made of ceramic.

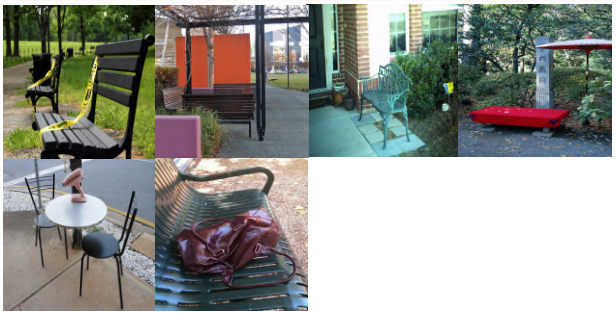
Task: Position **Scenario:** UIC **Model:** MiniCPM-V-2.6

Question: Is the umpire kneeling on the field in any of image 3, image 5 and image 6?

Ground Truth: yes

Model Response: no
Reasoning: in image 3, the catcher is standing and not kneeling. in image 5, the catcher is also standing. in image 6, the catcher is in a crouched position, not kneeling. therefore, the catcher is not kneeling in any of the images provided.

Task: Material **Scenario:** UIC **Model:** InternVL2.5-26B



Question: Is the bench wood in any of image 1, image 2, image 3, image 4 and image 6?

Ground Truth: no

Model Response: yes
Reasoning: images show toilets that are made of ceramic.
in image 1, image 2, and image 4, the benches are made of wood.
image 6 also shows a bench, but it is not clear if it is made of wood or not.

Task: Sentiment **Scenario:** UIC **Model:** GPT-4o

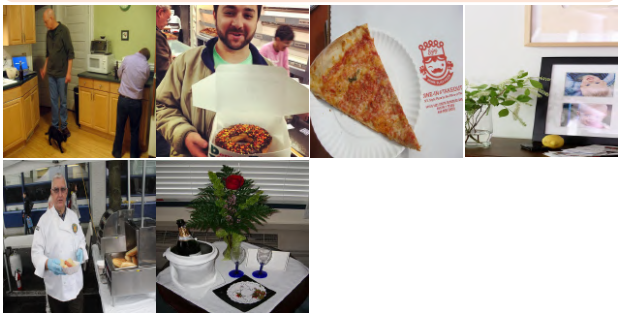
Question: Are people calm in any of image 1, image 3, image 4, image 5 and image 6?

Ground Truth: no

Model Response: yes
Reasoning: in images 1, 4, 5, and 6, people appear calm and composed, engaging in casual activities such as standing, sitting, or posing for photos. however, in image 3, the crowd seems more energetic and active, with people gathered closely and interacting, suggesting a more dynamic and less calm atmosphere.

Figure 15: Case study under UIC and DIC scenarios (Scaling contexts $k \geq 4$). denotes factual errors, denotes conflicts with target image, and denote correct target image details.

Task: Material **Scenario:** DIC **Model:** Qwen3-VL-8B



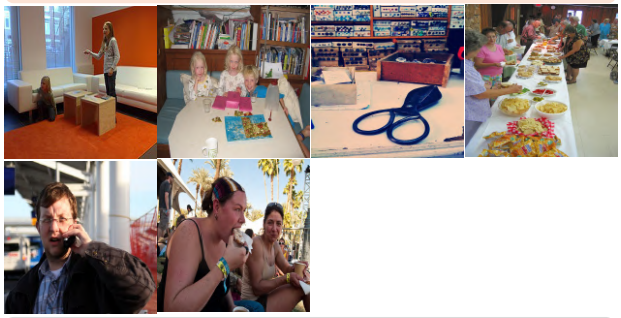
Question: Is the table glass in any of image 4 and image 6?

Ground Truth: no

Model Response: yes

Reasoning: in image 6, there are two wine glasses with blue stems placed on the table next to the bottle of champagne in the ice bucket. while image 4 shows a vase with flowers and a lemon, it does not contain any glassware. however, since the question asks if the table glass is in "any" of the images, and image 6 clearly contains table glasses, the answer is yes.

Task: Sentiment **Scenario:** DIC **Model:** InternVL-2.5-8B



Question: Is the lady looking happy or relaxed in any of image 1, image 4 and image 6?

Ground Truth: yes

Model Response: no

Reasoning: in image 1, the lady appears to be focused on playing a video game, which doesn't necessarily indicate happiness or relaxation. in image 4, the lady is eating a donut, which could suggest enjoyment, but it's not clear if she is happy or relaxed. in image 6, the lady is sitting and holding a cup, which might indicate relaxation, but it's not clear if she is happy.

Task: Sentiment **Scenario:** UIC **Model:** GPT-4o

Question: Is the man relaxed in any of image 1, image 3, image 5, image 6, image 7, image 8 and image 9?

Ground Truth: no

Model Response: yes

Reasoning: - **image 1:** the man appears to be in a thoughtful or pensive pose, with his hand resting on his face, which does not necessarily indicate relaxation. - **image 3:** the man walking along the train tracks with the sun setting behind him gives a sense of calm and freedom, suggesting relaxation. - **image 5:** the man sitting on the couch with his hands clasped and leaning forward appears more focused or serious than relaxed. - **image 6:** the man leaning on the church pew with a casual posture and hand resting on the bench suggests a relaxed demeanor. - **image 7:** the man on the phone appears engaged in a conversation, which does not necessarily indicate relaxation. - **image 8:** the man sitting on the fire hydrant with a casual posture suggests a relaxed state. - **image 9:** the man in the hallway with a loosened tie appears more tired or contemplative than relaxed.

Task: Object **Scenario:** UIC **Model:** GPT4o

Question: Is this a formal meeting in any of image 1, image 4, image 5, image 6 and image 7?

Ground Truth: no

Model Response: yes

Reasoning: in images 6 and 7, the settings and context suggest formal meetings.in image 6, the setting appears to be a formal dining event with a speaker addressing an audience in a formal environment.in image 7, the individuals are dressed in formal attire, seated at a table with name tags and formal dining arrangements, indicating a professional meeting or conference. in contrast, images 1, 4, and 5 appear to be more casual gatherings or social events, as indicated by the relaxed atmosphere, casual seating arrangements, and the presence of food and drinks.

Figure 16: Case study under UIC and DIC scenarios (Scaling contexts $k \geq 4$). denotes factual errors, denotes conflicts with target image, and denote correct target image details.